UNIVERSITÄT BAYREUTH

Master Thesis

Carbon use efficiency, biomass and activity of microorganisms in a thawing permafrost peat-land in Finnmark, Norway

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Abstract

The rising average global temperatures over recent decades are especially pronounced in the Arctic, where the thawing of permafrost soils could lead to positive climate feedback. Permafrost thawing can alter whole ecosystems and accelerate the release of soil organic carbon (C) into the atmosphere. Permafrost peatlands might react particularly sensitive to permafrost thaw as they represent globally a large C stock. However, knowledge on microbial efficiency and activity in permafrost peatlands is still limited. This study aimed to explore how thawing of a Norwegian palsa in form of active layer deepening and thermokarst affects the carbon use efficiency (CUE), biomass, growth and respiration of microorganisms. Peat cores from a palsa and a thermokarst feature in Áidejávri, Norway were sampled from six different depth sections and incubated under oxic and anoxic conditions to determine microbial parameters. Additionally, elemental concentrations and stoichiometric ratios were determined.

Mean microbial CUE was 0.37 in the palsa under oxic and 0.40 in the thermokarst under anoxic conditions across all depth sections, indicating no effect of oxygen availability and similar peat quality in these geomorphic features. In palsa and thermokarst, CUE of peat across all depth sections was positively correlated with hydrogen (H) concentration. In the palsa, CUE was, additionally, positively correlated with H/C ratio. This points to an increasing CUE with increasing degradability of organic matter. In the top depth sections, mean CUE reached values of up to 0.72, possibly due to a higher degradability of organic matter, but differences in CUE along the depth gradients were not significant. Microbial biomass C (MBC) across all depth sections was higher in the oxic palsa than in the anoxic thermokarst and presented rather low mean values of 525 µg C g peat^{-1} in the palsa and 252 µg C g peat^{-1} in the thermokarst. Similarly, microbial respiration across all depth sections displayed higher values in the oxic palsa with a mean of 44.0 µg CO_2-C g peat^{-1} d^{-1} and lower values in the anoxic thermokarst with a mean of 2.92 µg CO_2-C g peat^{-1} d^{-1}. Respiration was downregulated by anoxic conditions at both geomorphic features, and mean respiration rates were 12 to 45 times higher under oxic than under anoxic conditions. The results suggest that microbial activity at both geomorphic features is rather low due to strong degradation of the peat. Increasing intensity of permafrost thawing by rising temperatures may have little potential for positive climate feedback of these regional ecosystems.

Kurzfassung

Die steigenden globalen Durchschnittstemperaturen der letzten Jahrzehnte sind in der Arktis besonders ausgeprägt, wo das Auftauen von Permafrostböden eine positive Klima-Rückkopplung begünstigen könnte. Das Auftauen von Permafrost kann ganze Ökosysteme verändern und die Freisetzung von organischem Kohlenstoff (C) aus dem Boden in die Atmosphäre beschleunigen. Permafrostmoore könnten besonders empfindlich auf das Auftauen reagieren, da sie weltweit mit die höchsten C-Vorräte aufweisen. Bisher ist jedoch wenig über die Effizienz und Aktivität von Mikroorganismen in Permafrostmooren bekannt. Ziel dieser Studie war es daher, zu untersuchen, wie sich das Auftauen eines norwegischen Palsas durch die Vertiefung der aktiven Schicht oder durch Thermokarst auf die Kohlenstoffnutzungseffizienz (CUE), die Biomasse, das Wachstum und die Atmung von Mikroorganismen auswirkt. In Áidejávri in Norwegen wurden Torfkerne aus sechs verschiedenen Tiefenabschnitten eines Palsas und eines Thermokarsts entnommen. Diese wurden unter oxischen und anoxischen Bedingungen inkubiert, um mikrobielle Parameter zu bestimmen. Zusätzlich wurden Elementgehalte und stöchiometrische Verhältnisse bestimmt.

Die mittlere mikrobielle CUE betrug über alle Tiefen hinweg im Palsa 0,37 unter oxischen und im Thermokarst 0,40 unter anoxischen Bedingungen. Dies deutet darauf hin, dass die Sauerstoffverfügbarkeit keinen Einfluss auf die CUE hat und dass sich die Torfqualität zwischen Palsa und Thermokarst kaum unterscheidet. In Palsa und Thermokarst war die CUE über alle Tiefen hinweg positiv mit der Wasserstoff (H) -Konzentration und im Palsa zusätzlich positiv mit dem H/C-Verhältnis korreliert. Das deutet auf eine steigende CUE bei zunehmender Abbaubarkeit der organischen Substanz hin. In den oberen Torfschichten erreichte die mittlere CUE Werte von bis zu 0,72, was vermutlich in einer höheren Abbaubarkeit des organischen Materials begründet liegt. Die Unterschiede der CUE entlang der Tiefengradienten waren jedoch nicht signifikant. Der mikrobielle C (MBC) war über alle Tiefen hinweg im oxischen Palsa höher als im anoxischen Thermokarst und wies mit 525 µg C g Torf^{-1} im Palsa und 252 µg C g Torf^{-1} im Thermokarst eher niedrige Mittelwerte auf. Die mikrobielle Atmung wies über alle Tiefen hinweg ebenfalls im oxischen Palsa höhere Werte auf als im anoxischen Thermokarst. Hier lagen die durchschnittlichen Werte im Palsa bei 44,0 µg CO_2-C g Torf^{-1} d^{-1} und im Thermokarst bei 2,92 µg CO_2-C g Torf^{-1} d^{-1}. Des Weiteren wurde die Atmung sowohl im Palsa als auch im Thermokarst durch anoxische Bedingungen stark verringert und die mittleren Atmungsraten waren unter oxischen Bedingungen 12- bis 45-mal höher als unter anoxischen. Die Ergebnisse deuten auf eine eher geringe mikrobielle Aktivität in Palsa und Thermokarst hin, begründet durch den hohen Zersetzungsgrad des Torfes. Das fortschreitende Auftauen von Permafrostmooren durch steigende Temperaturen trägt daher vermutlich nur geringfügig zu einer positiven Klima-Rückkopplung dieser regionalen Ökosysteme bei.

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1 Introduction

Over the past 150 years there has been an increase in average global temperatures with the average global surface temperature being around 1.1 °C higher in the period 2011-2020 compared to the period 1850-1900 (IPCC, 2023). The permafrost zone is especially sensitive to climate change, and warming in these ecosystems was observed to be around twice as high as the global mean in the past decades (Hansen et al., 2007). This has already led to major impacts on permafrost landscapes, and further amplified permafrost thawing can be expected in the future (IPCC, 2023). As a result of low temperatures which reduce microbial activity and hinder the decomposition of plant material (Waldrop et al., 2010; Tao et al., 2023), permafrost soils represent large carbon (C) pools which are estimated at around 1,300 Pg C (Hugelius et al., 2014). The largest soil organic C stocks in the Northern Hemisphere were found in the top 100 cm (Tarnocai et al., 2009) or the top 300 cm (Hugelius et al., 2014) of permafrost peatland soils. Permafrost thaw will make organic C available for microbial decomposition (Song et al., 2014; Voigt et al., 2019) which could result in the release of C into the atmosphere in form of either carbon dioxide (CO_2) or, under anoxic conditions, methane (CH_4) with its larger global warming potential (Wilson et al., 2019; IPCC, 2023). Hence, permafrost thawing is likely one of the reasons for the increasing atmospheric CO_2 and CH_4 concentrations with a current global monthly mean concentration of 427 ppm CO_2 (as of May 2025; Lan et al., 2025a) and 1,935 ppb CH_4 (as of April 2025; Lan et al., 2025b). As permafrost peatlands with their large soil organic C stocks are common ecosystems in the Northern Hemisphere and cover up to 19% of the permafrost region (Tarnocai et al., 2009), their thawing could have a significant impact on global warming. However, the effect of thawing on the C pool, the fate of C in permafrost soils and finally on global warming is not fully understood yet (Schuur et al., 2008; McGuire et al., 2009; Schuur & Mack, 2018; Turetsky et al., 2020).

Permafrost soil is defined as soil that is frozen permanently, meaning for two or more consecutive years (van Everdingen, 1998). It can primarily be found in the subarctic, the Arctic (Brown et al., 1997), as well as in alpine regions (Bockheim & Munroe, 2014). Permafrost soils usually consist of a perennially frozen permafrost layer, a transition zone and a so-called active layer (Shur et al. 2005; Luo et al., 2025). Reaching temperatures above 0 °C in the growing season, the active layer is exposed to annual freeze-thaw-cycles, achieving its maximum depth in mid to late July (Tarnocai, 1980).

Permafrost peatlands started to accumulate organic matter (OM) since after the retreat of glacial ice, post ice-age (Zoltai, 1995; Routh et al., 2014; Kjellman et al., 2018). In Norway, peat age varies between ca. 5,000 and 10,000 cal. yr BP (Kjellman et al., 2018). Permafrost aggregation started between 100 and 950 cal. yr BP, indicating that most permafrost peatlands in Norway developed epigenetically. Thick permafrost layers combined with an insulating snow

cover and anoxic conditions in the peat can result in the formation of palsas (Seppälä, 1986; Gurney, 2001). Palsas are dome-shaped, upheaved peat structures with a permanently frozen core (Zoltai & Tarnocai, 1975; Seppälä, 1986) and are usually dominated by vascular plants (Kjellman et al., 2018; Harris et al., 2023). In the growing season, the thawed active layer of palsas can become relatively dry and present oxic conditions because of drainage and an exposed surface resulting from the upheaval (Laberge & Payette, 1995; Lawrence et al., 2015). Therefore, there can be parts of palsas which do not grow any plants (Martin et al., 2019).

In contrast to palsas with their intact permafrost structures, consisting of active and permafrost layer, thermokarsts can appear when palsas collapse due to gradual deepening of the active layer (Jorgenson & Osterkamp, 2005; Kokelj & Jorgenson, 2013). Thermokarsts usually display high water saturation and anoxic conditions (Jorgenson & Osterkamp, 2005; Connon et al., 2014; Zandt et al., 2020). After collapse, the former palsa peat is overgrown by Sphagnum mosses (Kjellman et al., 2018; Harris et al., 2023), and a new layer of peat accumulates as the thermokarst matures (Wilson et al., 2017; Taylor et al., 2019; Harris et al., 2023). Consequently, permafrost thaw in permafrost peatlands as a result of climate warming can occur either in form of active layer deepening or through resulting thermokarst (Olefeldt et al., 2016).

Microorganisms play an important role for the stabilization of organic C in soils. Organic C can be stored through microbial growth either in the living microbial biomass or in form of microbial residues which have a high recalcitrance and, hence, can stabilize C over long periods of time (Gunina & Kuzyakov, 2015). On the other hand, microbes also respire C, usually in form of CO_2, to gain energy which will result in a loss of C from OM (Iven et al., 2022). The C use efficiency (CUE) is the ratio of C that is invested in microbial growth and C that is respired by microorganisms, and it is a commonly used parameter for the amount of microbial C stored in soil (del Giorgi & Cole, 1998; Manzoni et al., 2012). Typically, microbial communities in soils display an average CUE of around 0.55 (Sinsabaugh et al., 2013; Jones et al., 2018). Jones et al. (2018) documented CUEs ranging between 0.28 and 0.78 across 114 different soils and 16 different substrates. A high CUE value indicates that a larger portion of available C is invested in growth and remains in the soil in form of microbial biomass or necromass while a smaller portion of the available C is respired and lost to the atmosphere (Manzoni et al., 2012). Hence, it can be expected that microbial CUE and soil organic C stock are positively correlated (Tao et al., 2023; Duan et al., 2024; Tao et al., 2024). In peatlands, however, most C consists of hardly decomposed plant litter, protected through anoxic conditions and chemical recalcitrance (Sjörs, 1980; Rydin & Jeglum, 2013). The proportion of microbial derived necromass in the C stock of permafrost peatlands has not yet been sufficiently researched. Accelerated microbial decomposition of peat following permafrost thawing could potentially influence microbial growth and, thus, the proportion of microbial necromass.

CUE is affected by various factors such as nitrogen (N) availability, C/N ratio, availability of easily degradable C as well as oxygen (O_2) availability. N limitation triggers a higher production of extracellular enzymes, which will result in a lowered CUE (Sinsabaugh et al., 2013) since energy might be spilled (Russell & Cook, 1995). Adding to that, a low N availability can provoke N mining, a process where decomposition of labile substrates is enhanced to access N from recalcitrant sources (Craine et al., 2007). This can lower the CUE as microbial respiration is accelerated (Manzoni et al., 2012). Concurrently, a decreasing N availability might trigger a shift in microbial community structure to less efficient communities that show higher respiration rates (Ågren et al., 2001). Moreover, CUE can be affected by the C/N ratio (Manzoni & Porporato, 2009; Manzoni et al., 2012; Sinsabaugh et al., 2013; Sinsabaugh et al., 2016; Fang et al., 2020). Microbes can react to high C/N ratios with an increased respiration, named overflow respiration, in order to get rid of the excessive C and achieve a ratio that matches the C/N ratio of the organism (Manzoni & Porporato, 2009). The availability of easily degradable C can, additionally, lead to changes of the CUE and can be assessed e.g. with the ratio of hydrogen (H) to C (Zaccone et al., 2018). C limitation and the decomposition of recalcitrant C is associated with an increase of microbial respiration and slower microbial growth (Blagodatskaya & Kuzyakov, 2008; Manzoni et al., 2012; Öquist et al., 2017) since more energy is being invested in enzyme production (Manzoni et al., 2012). Furthermore, studies state that CUEs decline under anoxic conditions as compared to oxic conditions (Šantrůčková et al., 2004; Manzoni et al., 2012; Zheng et al., 2019) because of a change of metabolic end products. As a response to O_2 limiting conditions, CH_4 becomes a more common metabolite (Burgin et al., 2011). Since CH_4 cannot be completely oxidized, the energy gain for microorganisms is lower, and microbial growth declines (Manzoni et al., 2012; Zheng et al., 2019). However, so far it is unknown how these factors affect the CUE in thawing permafrost peatlands.

Peat from mature thermokarsts is expected to be depleted in N and show a higher C/N ratio and a lower content of easily degradable C than peat from palsas. Chen et al. (2018) observed a decreasing CUE in a permafrost swamp meadow on the Tibetan Plateau after 16 years of permafrost collapse and explained it with a decreasing N availability. Thawing will lead to a release of nutrients, making them available for plant and microbial use (Reyes & Lougheed, 2015; Finger et al., 2016), and these nutrients will possibly be depleted as the thermokarst matures (Chen et al., 2018). Furthermore, the C/N ratio of thermokarsts was observed to be slightly higher than that of palsas (Pengerud et al., 2013; Chen et al., 2018; Kirkwood et al., 2021). Additionally, peat from intact permafrost layers of palsas is being preserved under the frozen conditions (Zimov et al., 2006; Waldrop et al., 2010) while thawing with thermokarst accelerates decomposition in peat (Harden et al., 2012; Hodgkins et al., 2014). Over time of permafrost collapse, enhanced decomposition was proposed to result in the depletion of the easily degradable C pool (Hodgkins et al., 2014; Semenchuck et al., 2016; Monteux et al.,

2018). These observations could point to a higher CUE in palsa peat than in thermokarst peat. Microbial respiration was found to be restricted by the in-situ anoxic conditions in thermokarsts compared to the in-situ oxic conditions of palsas (Bäckstrand et al., 2010; Estop-Aragonés et al., 2018b). On the other hand, microbial biomass and activity (i.e. microbial growth and respiration) in permafrost peat from palsas is restricted by the long-term freezing conditions (Jansson & Taş 2014; Altshuler et al., 2017), and it was found that under the same O_2 conditions, potential respiration of permafrost peat shortly after thawing is lower than in thermokarst peat (Turetsky, 2004). In more long-term incubation studies, it was lower (Kirkwood et al., 2021) or similar (Estop-Aragonés et al., 2022; Harris et al., 2023) to that of thermokarst peat. Similar tendencies can also be expected for microbial biomass C (MBC) and growth (Waldrop et al., 2010; Song et al., 2014). In consequence, CUE, MBC, growth and respiration are possibly higher in oxic palsa peat than in anoxic thermokarst peat, but under the same O_2 conditions, CUE is likely higher in palsa peat while MBC, microbial growth and respiration are probably lower than in equivalent thermokarst peat.

Furthermore, the active layer peat of palsas possibly presents a larger N limitation and higher C/N ratios than peat of the permafrost layer. Comparable to thermokarst, thawing of the active layer will release nutrients and make them available for use to plants and microbes (Keuper et al., 2012; Reyes & Lougheed, 2015; Finger et al., 2016). Combined with the oxic conditions, mineralization in the active layer might be enhanced during the growing season which could facilitate the leaching of these nutrients (Updegraff et al., 1995; Freppaz et al., 2007; Keuper et al., 2012; Reyes & Lougheed, 2015). At the same time, plant litter accumulates at the top layer of palsas, and studies have found higher C/N ratios in the active layer of permafrost peatlands compared to the permafrost layer (Song et al., 2014; Kjær et al., 2024). This could point to a higher degradability of substrates in the active layer (Malmer & Holm, 1984; Chen et al., 2003; Zaccone et al., 2018) which could enhance CUE (Blagodatskaya & Kuzyakov, 2008; Manzoni et al., 2012; Öquist et al., 2017), but the imbalanced C/N ratios mainly suggest a lower CUE in the top active layer (Manzoni & Porporato, 2009; Manzoni et al., 2012; Sinsabaugh et al., 2013; Sinsabaugh et al., 2016; Fang et al., 2020). Microbial biomass and activity are possibly restricted by the frozen conditions in permafrost layers (Jansson & Taş 2014; Altshuler et al., 2017) and enhanced by a higher O_2 availability (Fisk et al., 2003) and a constant substrate input to top active layers (Fisk et al., 2003; Walz et al., 2017; Kjær et al., 2024). Microbial biomass and respiration were observed to be highest in the top active layer of palsas even after laboratory incubation at higher temperatures and stable O_2 conditions (Song et al., 2014; Kirkwood et al., 2021; Kjær et al., 2024). Therefore, CUE is likely lower in the active layer than in the permafrost layer of palsas, while microbial biomass and activity should show an opposite pattern.

In thermokarsts, the newly formed peat possibly displays a lower availability of N and a higher C/N ratio than peat from deeper depth sections (former palsa peat). The new thermokarst peat is dominated by Sphagnum mosses and, hence, to a majority affected by Sphagnum litter whereases the former palsa peat is primarily made up of vascular plant litter (Kjellman et al., 2018; Harris et al., 2023). Thus, N limitation might be higher in the newly formed thermokarst peat because Sphagnum can use N from the peat in a more efficient way than vascular plants, leading to a more profound N depletion (Malmer et al., 2003). Moreover, the C/N ratio in newly formed thermokarst peat at the top was observed to be especially high compared to other depth sections (Harris et al., 2023), and Oke and Hager (2020) reported a higher C/N ratio in Sphagnum peat compared to vascular plant peat. This could propose a lower CUE in the top, newly formed thermokarst peat than in the deeper depth sections. In contrast, even after incubating under the same conditions, respiration of new thermokarst peat was found to be higher than that of deeper depth sections (Harris et al., 2023; Kjær et al., 2024), possibly because of a constant substrate input (Fisk et al., 2003; Harris et al., 2023; Kjær et al., 2024). This is why CUE is probably lower in the newly formed thermokarst peat, but microbial biomass and activity are likely higher than in the former palsa peat.

This research aims to study the effect of thawing, i.e. active layer deepening and thermokarst on the fate of microbial C use in permafrost peatlands. It is hypothesized that:

1) (a) Under different oxygen conditions, oxic palsa peat exhibits higher microbial CUE, biomass, growth and respiration than anoxic thermokarst peat, but (b) under same oxygen conditions (oxic and anoxic) palsa peat exhibits higher microbial CUE and lower microbial biomass, growth and respiration than thermokarst peat.
2) (a) In oxic palsa peat, microbial CUE is lower, and microbial biomass, growth and respiration are higher in the active layer than in the permafrost layer and (b) in anoxic thermokarst peat, microbial CUE is lower, and microbial biomass, growth and respiration are higher in the newly formed peat (top depth section) than in the deeper depth sections.
3) Microbial CUE, biomass, growth and respiration in peat from palsa and thermokarst are higher under oxic than under anoxic conditions.

To investigate these hypotheses, the CUEs of a Norwegian palsa and a nearby mature thermokarst were determined in different depth sections by estimating MBC, potential microbial growth and respiration. The samples were incubated under oxic and anoxic conditions to allow for comparability between the two geomorphic features and also estimate the effect of O_2 availability. Moreover, elemental concentrations and stoichiometric ratios of the peat samples were determined to disclose possible connections between stoichiometry, permafrost thaw and the assessed microbial parameters.

2 Materials and methods

2.1 Study site

The peat was collected from thermokarst hollows (AT) and palsa plateaus (AP) in Áidejávri in Finnmark, Norway (Fig. 1). The site is located at approximately 68.75 N and 23.33 E. Áidejávri is placed on a plateau called Finnmarksvidda with an area of ca. 22,000 km^2 and at an elevation of around 370 m above sea level (Kjær et al., 2024). The mean annual temperature reported from the nearest weather station (Šihččajávri) is around -1.6 °C, and a mean annual precipitation of 577 mm was documented (reference period 1991-2020; Norwegian Centre for Climate Services, n.d.). The study site is a permafrost peatland, consisting of intact palsas as well as thermokarsts with different successional stages. According to Kjær et al. (2024) the soil in Áidejávri can be characterized as a Histosol (IUSS Working Group WRB, 2022), and the exact age of the peatland in Áidejávri is unknown (Kjær et al., 2024). Throughout the whole time before permafrost started to form, the peatland was most likely a fen, as evidenced by macrofossil data (Henrich, 2025). It was assumed that the thermokarst structures in Áidejávri developed within the last century, supposing a similar accumulation rate of C as nearby thermokarsts (Henrich, 2025).

Samples were collected from two thermokarst hollows and a surrounding palsa (Fig. 1c). Based on macrofossil data, the thermokarst cores AT2, AT3 and AT5 were sampled from a thermokarst hollow that is dominated by *Sphagnum lindbergii* Schimp. as well as some *Eriophorum* L. sp. (Henrich, 2025). The dominating vegetation at the palsa consisted of *Empetrum nigrum* L., *Ledum palustre* L., *Vaccinium uliginosum* L., *Rubus chamaemorus* L., and various lichens such as *Cladonia* P. Brown spp. as well as mosses, but there was only little to no Sphagnum. Additionally, *Betula nana* L., *Andromeda polifolia* L. and *Vaccinium vitis-ideae* L. were present at the palsa.

2.2 Peat sampling

Four palsa cores and four thermokarst cores were sampled in mid-July of 2024 (Fig. 1c-e). All four thermokarst cores were taken from mature thermokarsts. The successional stages of the thermokarst were determined based on differences in physical features and the present vegetation (Henrich, 2025). Three of the thermokarst cores were collected from the same thermokarst hollow (AT2, AT3 and AT5) while one was collected from another hollow (AT1; Fig. 1c). The palsa cores were sampled close to the thermokarst hollows (Fig. 1c). The cores were divided into five different depth sections, respectively (Tab. 1). The samples of all palsa cores were taken from the assigned depth sections 1 to 5. The absolute sampling depths of the thermokarst cores were chosen in a way that allows comparability between palsa and

thermokarst. Since permafrost collapse caused the peat to drop, it was individually determined, based on peat morphology, which absolute depths of each thermokarst core were equivalent to the sampled depths of the palsa. Hence, although having different absolute depths (Tab. 1), depth section 1 of each thermokarst core was expected to be the same age as depth section 1 of the palsa cores and so on. However, for depth section 3 no corresponding samples were taken from the thermokarst cores. Moreover, the newly formed peat of the thermokarst was assigned to depth section 0 which represents the top 40 cm of all thermokarst cores (Tab. 1).

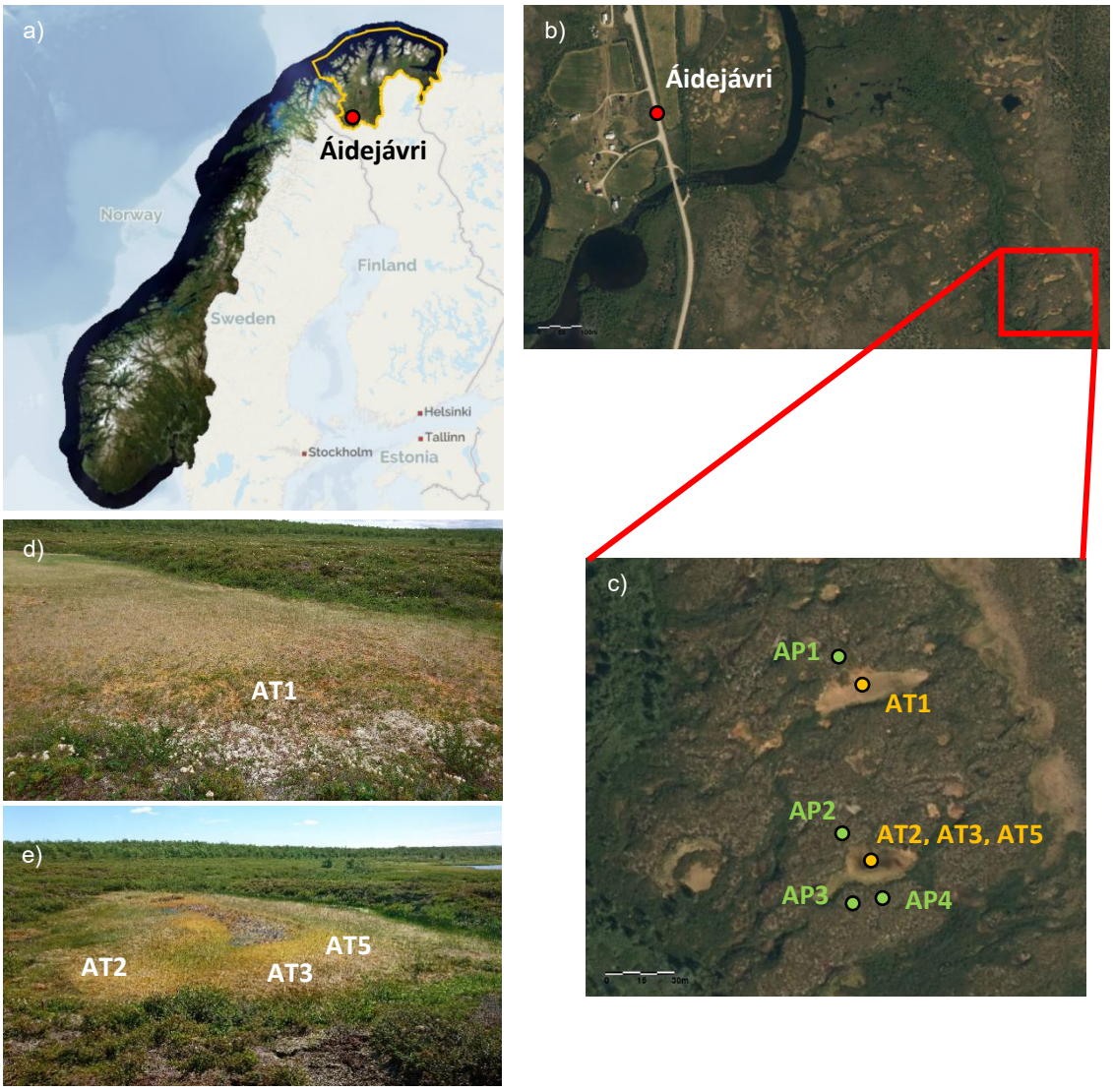


Figure 1: *a) Norway with Finnmark in yellow outline and location of Áidejávri as well as b, c) aerial images of Áidejávri, including the locations of the sampling points from palsa (AP) and thermokarst (AT; Norwegian Mapping Authority, n.d.), and d, e) photographs of the thermokarst cores. The photographs were taken by Arwen Henrich in 2024 during peat sampling.*

At the end of August 2024, the thawing depth of the palsa was at around 70 cm (Henrich, 2025). Hence, depth sections 1, 2 and 3 of the palsa will be interpreted as the active layer which shows annual freeze-thaw cycles (Tab. 1). Depth sections 4 and 5 will be interpreted as permafrost peat which stays frozen throughout the year.

The sampling of the first 40 cm of the thermokarst cores was performed manually with a bended aluminum sheet (50x6x13 cm). The deeper parts of the cores were collected with a Russian peat corer (5.2 cm diameter; Royal Eijelkamp, Giesbeck, Netherlands). In the palsa, the samples from the active layer were cut out with a serrated knife while the deeper depths from the permafrost layer were sampled by hammering a steel pipe (3 cm diameter) into the frozen ground. After sampling, the peat was kept frozen at -24 °C until the beginning of the laboratory analyses in November 2024. Only during the transport from Norway to Bayreuth, the freezing temperature was less than -24 °C.

Table 1: *Absolute sampling depths of the six depth sections (0-5) at four plots of a Norwegian palsa (AP1-4) and four plots of a Norwegian thermokarst (AT1-3, AT5).*

Geomorphic feature	**Plot**	**Depth section**					
		Newly formed peat	**Active layer/ former active layer**			**Permafrost layer/ former permafrost layer**	
		0 (cm)	**1** (cm)	**2** (cm)	**3** (cm)	**4** (cm)	**5** (cm)
Palsa	AP1 - 4	-	0-20	20-40	50-70	80-100	100-120
Thermokarst	AT1	0-40	63-83	83-103	-	143-163	163-183
Thermokarst	AT2	0-40	43-63	63-83	-	123-143	143-163
Thermokarst	AT3	0-40	60-80	80-100	-	140-160	160-180
Thermokarst	AT5	0-40	50-70	70-90	-	130-150	150-170

2.3 Sample preparation

Prior to the laboratory analyses, the frozen samples were thawed at 5 °C over night in the beginning of November 2024. After thawing, all disruptive materials such as roots or insects were removed, and the samples were homogenized with a mixer. Whenever no analyzes were performed, the thawed and homogenized samples were stored at 2 °C. All samples, except four, were considered water saturated. For the remaining four samples, the maximum water holding capacity (WHC_{max}) was determined. Around ¼ of glass-fritted drainage tubes was filled with peat sample and weighed. Next, VE water was added to the samples. The glass-fritted drainage tubes were closed with parafilm and stored at 5 °C until all of the excess water was

drained off and the peat samples had achieved their WHC_{max}. The samples were weighed, dried at 105 °C for 24 hours and weighed again. In the following, the WHC_{max} in g g peat^{-1} was calculated with equation (1):

$$\mathrm{WHC_{max}} = \frac{(\mathrm{fw_{WHC_{max}}} - \mathrm{dw})}{\mathrm{dw}} \quad (1)$$

Where $\mathrm{fw_{WHC_{max}}}$ is the fresh weight of the peat at WHC_{max} (g) and dw is the dry weight of the peat (g).

Finally, the gravimetric water content (g g peat^{-1}) for all samples was calculated in the same way as the water holding capacity (eq. 1) by weighing the original peat samples before and after drying the peat at 105 °C for 24 hours.

2.4 Determining elemental concentrations and stoichiometric ratios

Elemental concentrations and stoichiometric ratios of the samples were determined with the dry combustion method (Matejovic, 1993; Matejovic, 1997). For this, ca. 5 g of peat sample were dried at 40 °C for three days. The dried samples were then ground (MM 400, Retsch GmbH, Haan, Germany). Next, 15 mg of the ground samples and ca. 45 mg of tungstic oxide powder were weighed into tin boats, folded and analyzed for C, H, N and sulfur (S) (vario EL cube, Elementar Analysesysteme GmbH, Langenselbold, Germany). For the O analyzes, 2-3 mg of ground sample were weighed into silver boats, folded and measured (rapid OXY cube, Elementar Analysesysteme GmbH, Langenselbold, Germany). The derived results were converted into mg g peat^{-1}, and stoichiometric ratios were calculated mole-based.

2.5 Microbial respiration

In a first step, the peat samples were air-dried to achieve a homogenous water content of 150% throughout all the samples (250% for depth section 1 of the thermokarst samples as those had a very high WHC_{max}) and then prepared for the pre-incubation. Ca. 15 mg of all palsa samples as well as the thermokarst samples of depth sections 4 and 5 were pre-incubated under oxic conditions for nine days at 15 °C, respectively. Accordingly, ca. 15 mg of all thermokarst samples as well as the palsa samples of depth sections 4 and 5 were pre-incubated under anoxic conditions, respectively, by flushing the incubation flasks with N_2 for 30 minutes. The pre-incubation of the anoxic samples lasted for 20 days at 15 °C.

After finishing pre-incubation, the microbial respiration was determined. For this, the incubation flasks of the oxic samples were flushed with synthetic air for 15 minutes, and an overpressure was created. In the following, the CO_2 production was determined by measuring the concentration of CO_2 four times (t0, t1, t2, t3) over a period of 48 hours with a gas chromatograph (8610C GC, SRI Instruments Europe GmbH, Bad Honnef, Germany). At t3, the CH_4 concentration was measured, additionally. The incubation flasks of the anoxic samples were flushed with N_2 for 20 minutes, and overpressure was created. Since the anoxic conditions were expected to support the production of CH_4 rather than CO_2 (Burgin et al., 2011; Wilson et al., 2019), a measurement of both CO_2 and CH_4 concentration for four times (t0, t1, t2, t3) over a period of 160 hours followed. However, since no CH_4 production was measured for both the oxic and the anoxic samples (see chapter 3.4), respiration was calculated only based on the values of the measured CO_2 production. The produced CO_2 – C over the measuring period ($\Delta CO_2 - C$) in µg was calculated with equation (2):

$$\Delta CO_2 - C = \frac{CO_2(ppm) \times V_F \times P}{R \times T} \times M \quad (2)$$

Where CO_2(ppm) is the produced CO_2 over the time of incubation (ppm), calculated by multiplying the slope of the function representing the increase of CO_2 over time (ppm h^{-1}) with the time of incubation (h), V_F is the volume of the incubation flask reduced by the volume of sample inside the flask (m^{-3}), P is the total pressure, meaning the sum of the overpressure in the incubation flask and the surrounding air pressure (Pa), R is the gas constant of 8.314 (J K^{-1} mol^{-1}), T is the incubation temperature (K) and M is the molar mass of C (g mol^{-1}).

The C invested in respiration ($C_{Respiration}$) in µg CO_2-C g $peat^{-1}$ d^{-1} was then calculated using equation (3):

$$C_{Respiration} = \frac{\Delta CO_2 - C}{dw \times \frac{24}{t}} \quad (3)$$

Where dw is the dry weight of the peat sample (g) and t is the time of incubation (h).

2.6 Microbial biomass carbon

In a next step, the MBC was determined by applying the chloroform fumigation-extraction (CFE) method (Vance et al., 1987). The aim of the method is to calculate the MBC by determining the difference between the total organic C (TOC) that was extracted from a soil fumigated with chloroform and a replicate of the same soil sample that was left unfumigated. Since

the fumigation with chloroform leads to lysis of the microbial cells in the samples, the difference between TOC in fumigated and unfumigated soils, corrected with a conversion factor, is expected to equal the MBC (Vance et al., 1987). For this, ca. 5 g of each respective sample were weighed into two jars, respectively. One half of the replicates was extracted immediately. Extraction was performed by adding 25 ml of a 0.5M K_2SO_4 solution to the samples, respectively (Horwath & Paul, 1994). In the following, the samples were shaken for 1 hour at 12 $\frac{1}{\text{min}}$ (Reax 20, Heidolph Scientific Products GmbH, Schwabach, Germany) and finally filtered (Rotilabo®-vouwfilters, Type 113P, cellulose, membrane diameter 125 mm, Carl Roth GmbH+Co. KG, Karlsruhe, Germany).

The other half of the replicates was fumigated prior to extraction. The samples were put in a desiccator, chloroform was added in an open jar, and the desiccator was sealed. Next, the desiccator was evacuated by creating a vacuum which caused the chloroform to boil and, hence, exposed the peat samples to the chloroform vapor. The evacuation lasted for 5 minutes to ensure that the chloroform vapor reached the peat samples completely. After 48 hours, the chloroform was removed by evacuating the desiccator and re-opening it to the atmosphere for 5 times. The samples were left open for another hour outside of the desiccator to get rid of all the remaining chloroform before adding the K_2SO_4 and performing the extraction in the same way as described above for the unfumigated samples.

After extraction, the derived solutions were stored at -20 °C and, in a next step, all samples were analyzed for their TOC content. The TOC values were converted into µg C g peat^{-1}, and MBC in µg C g peat^{-1} was calculated with equation (4):

$$\text{MBC} = \frac{\text{F}_\text{C}}{\text{k}_\text{C}} \tag{4}$$

Here, F_C is (organic C extracted from fumigated soil (µg C g peat^{-1})) – (organic C extracted from non-fumigated soil (µg C g peat^{-1})) (Wu et al., 1990), and k_C is 0.45, a conversion factor which reflects the proportion of C that can be extracted from the soil after fumigation (Joergensen, 1996).

However, some of the calculated values were negative (Tab. A.1). This might be a result of methodological difficulties, stemming from the high organic C content in the peat and the high variability within the samples (see chapter 4.5.1). To handle the negative values and make the following calculation of the CUE possible, a minimum value of 50.0 µg C g peat^{-1} was set (see chapter 4.5.1), and all negative MBC values as well as the values below 50.0 µg C g peat^{-1} were set to this minimum value (Tab. A.1).

2.7 Microbial carbon use efficiency

Finally, the ^{18}O-method was applied to determine microbial growth and, finally, the microbial CUE (Spohn et al., 2016a; Zheng et al., 2019). This method is based on adding $H_2{}^{18}O$ to soil samples and incubating them for about 48 hours while $H_2{}^{16}O$ is added to sample replicates. After incubation, the DNA of all samples is extracted, and by analyzing the at% of ^{18}O in the labelled samples ($H_2{}^{18}O$) and in the natural abundance control samples ($H_2{}^{16}O$), the production of DNA during the incubation time can be determined and later converted into microbial growth (Spohn et al., 2016a).

For this, 300 mg of each pre-incubated peat sample were weighed into vials two times. 150 µl of $H_2{}^{18}O$ were added to one of the replicates to achieve an ^{18}O enrichment of ca. 41.0 at%, and 150 µl of $H_2{}^{16}O$ were added to the second replicate to prepare the natural abundance controls. For the anoxic samples the weighing was done prior to the anoxic pre-incubation (Chapter 2.5) into air-tight glass vessels, and the $H_2{}^{18}O$ and $H_2{}^{16}O$ were added with a syringe to ensure that the samples did not get in contact with O_2. Since the syringe was not as well calibrated as the pipette that was used for the oxic samples, the added $H_2{}^{18}O$ or $H_2{}^{16}O$ was weighed to avoid any inaccuracies in the calculations. To allow an incorporation of the ^{18}O into the microbial DNA, the samples were then incubated for ca. 48 (oxic) or ca. 64 (anoxic) hours at 15 °C, respectively. The $H_2{}^{16}O$ samples were incubated in the same way as the $H_2{}^{18}O$ samples.

After incubation, DNA extraction of both the $H_2{}^{18}O$ and $H_2{}^{16}O$ samples was performed using the SPINeasy™ DNA Kit for Plant (MP Biomedicals). The DNA was extracted using the provided solutions and according to the protocol of the kit. All needed equipment, including glasses, microcentrifuge tubes and pipette tips were sterilized before use. 200 µl of an equilibration buffer were used to clean the filters of the columns. Next, 900 µl of a lysis buffer and 15 µl of RNase A were added to the prepared 300 mg of peat sample. The samples were homogenized by bead-beating and, in the following, centrifuged for 10 minutes at 14,000 g. 700 µl of the supernatant were transferred into a microcentrifuge tube. However, for most of the samples it was not possible to take exactly 700 µl of supernatant. Since the peat samples were dried to a water content of 150% but had far higher water holding capacities, a large part of the added lysis buffer was soaked up by the peat, and the peat had to be pressed down to gain as much liquid as possible and approximately achieve the intended 700 µl. For the samples 17, 18, 23 and 24 an additional amount of 300 µl lysis buffer had to be added and centrifuged again to gain enough supernatant. In the following, 200 µl of a PPS solution were added, and the microcentrifuge tubes were centrifuged for 1 minute at 14,000 g. 700 µl of the resulting supernatant were then transferred to a new microcentrifuge tube, 700 µl of binding buffer were added, and everything was mixed by pipetting up and down. 700 µl of this mixture were then

added to the previously prepared filter columns with collection tubes and centrifuged for 1 minute at 14,000 g. After discarding the flowthrough, the step was repeated. Next, 750 µl of wash buffer were added to the filter columns which were then centrifuged for 1 minute at 14,000 g. Again, the step was repeated after the flowthrough was discarded. In the following, the filter columns were centrifuged for an additional minute at 14,000 g and incubated at 55 °C for 5 minutes. The collection tube was removed from the filter column, and the filter column was placed onto a new, pre-weight microcentrifuge tube. 100 µl of elution buffer were added, and everything was incubated for 5 minutes at room temperature. Then, the filter columns with microcentrifuge tubes were centrifuged for 1 minute at 12,000 g, and the eluted DNA was collected in the microcentrifuge tubes which were weighed and immediately frozen to -80 °C.

An aliquot of the eluate was used to quantify the extracted DNA. The quantification was performed by using PicoGreen (Sandaa et al., 1998). 5 µl of eluted DNA as well as 50 µl of PicoGreen (Thermo Fisher Scientific) and 45 µl of 20x TE buffer (200mM TRIS HCl, 20 mM EDTA; Thermo Fisher Scientific) were added to wells of a 96-well microplate. Additionally, a 96-well microplate for a calibration curve was prepared by adding different concentrations of λDNA (0, 6.2 and 25 µl; Thermo Fisher Scientific), a corresponding amount of TE buffer to reach 50 µl in total (50, 43.8 and 25 µl), and 50 µl of PicoGreen. The 96-well microplates were placed in a microplate reader (Infinite M200 Pro, Tecan Trading AG, Männedorf, Switzerland), and the fluorescence was measured at 480 nm excitation and 520 nm emission wavelength (Sandaa et al., 1998). The gain was set to optimal (100%) for the calibration plate, and for the sample plate the gain was manually set to the optimal gain determined for the calibration plate. The slope of the calibration curve was then used to convert the values of fluorescence reading into DNA concentrations.

In a next step, the extracted DNA was purified with the GeneClean® II kit (MP Biomedicals). The purification was performed with the provided solutions and followed the protocol. 70 µl of eluted DNA were transferred to microcentrifuge tubes and 210 µl of NaI solution as well as 5 µl of glassmilk were added. Everything was mixed by pipetting up and down, and the mixture was incubated for five minutes at room temperature before being centrifuged for 30 seconds at 14,000 g. Next, the supernatant was manually discarded. 500 µl of prepared New Wash were added to the centrifuge tubes and everything was mixed by pipetting up and down. The mixture was centrifuged for 30 seconds at 14,000 g, and the supernatant was discarded manually. Again, 500 µl New Wash were added, resuspended, centrifuged, and the supernatant was discarded. The caps of the microcentrifuge tubes were then left open at room temperature until the pellets of glassmilk and DNA were completely dry. After drying, 5 µl of water were added to the microcentrifuge tubes, the mixture was resuspended by pipetting up and down and then centrifuged for 30 seconds at 14,000 g. The supernatant containing the purified,

eluted DNA was transferred into a new, pre-weight microcentrifuge tube which was again weighed after adding the eluted DNA. Twelve of the purified samples were again tested for their DNA quantity to determine the loss of DNA during the purification step (Tab. A.2). After purification, the samples were stored at -20 °C and later pipetted into silver capsules and dried at 55 °C over night. Finally, the $^{18}O/^{16}O$ isotope ratio of the DNA samples was analyzed by the Competence Centre for Stable Isotopes of the Georg-August-University Göttingen with the use of an IRMS.

Based on the determined values of microbial respiration, MBC and ^{18}O abundance, the CUE was calculated according to equation (5) (Manzoni et al., 2012):

$$\mathrm{CUE} = \frac{\mathrm{C_{Growth}}}{(\mathrm{C_{Growth}} + \mathrm{C_{Respiration}})} \tag{5}$$

Where $\mathrm{C_{Growth}}$ is the microbial growth in µg C g peat^{-1} d^{-1} that was calculated using the amount of produced DNA during the incubation (DNA_p) in µg (Zheng et al., 2019; Tian et al., 2023):

$$\mathrm{C_{Growth}} = \frac{\mathrm{DNA_P} \times \mathrm{MBC}}{\mathrm{DNA_c} \times \mathrm{dw} \times \frac{24}{\mathrm{t}}} \tag{6}$$

with

$$\mathrm{DNA_P} = \mathrm{DNA_O} \times \left(\frac{18\mathrm{O_{ex}}}{18\mathrm{O_{en}}}\right) \times \left(\frac{100}{\mathrm{DNA_{O_2}}}\right) \tag{7}$$

where MBC is the microbial biomass C (µg g peat^{-1}) (eq. 4), DNA_c is the determined DNA concentration (µg g peat^{-1}), dw is the dry weight of the peat (g), t is the time of incubation (h), DNA_O is the measured amount of O from the isotopic analyses (µg), $18O_{ex}$ is the excess of ^{18}O in the samples compared to the ^{16}O replicates (at%), $18O_{en}$ is the enrichment of ^{18}O within the respective sample, depending on the amount of ^{18}O added to the sample before DNA extractions (at%), and $\mathrm{DNA_{O_2}}$ is the O_2 content of DNA (weight%).

$\mathrm{C_{Respiration}}$ was calculated based on equations (2) and (3). Additionally, the turnover time of microbial biomass in days was calculated using equation (8) (Zheng et al., 2019; Tian et al., 2023):

$$\mathrm{Turnover\ time} = \frac{\mathrm{DNA}}{\mathrm{DNA_p} \times \frac{24}{\mathrm{t}}} \tag{8}$$

where DNA is the quantity of DNA (µg), DNA_P is the produced DNA during the incubation (µg), calculated according to equation (7) and t is the time of incubation (h).

The laboratory analyses, including respiration measurements, determination of MBC as well as DNA extraction, quantification and purification were performed within one month after starting the pre-incubation, and the samples were stored at 15 °C throughout the whole time to ensure comparability of the derived results for the final CUE calculation.

2.8 Statistical analysis

Calculations were performed in Microsoft Excel (version 2108; Microsoft Corporation, 2024). Plots were created and all statistical analyzes were performed in R (version 4.2.3; R Core Team, 2023). Prior to all of the main statistical tests, groups and residuals were tested for normality and homogeneity of variances. Normality of data was tested using the Shapiro-Wilk test as well as descriptive statistics by creating qqplots and histograms. Homogeneity of variances was tested using Levene's test from the "car" package (Fox & Weisberg, 2019) as well as descriptive statistics by plotting the fitted data vs. residuals. To achieve the best data distribution throughout all the applied tests, CUE, MBC and microbial turnover time were transformed using the square root, and respiration and growth were transformed using the natural logarithm. C/N ratio was transformed using the Yeo-Johnson transformation (Yeo & Johnson, 2000). For the other elemental concentrations and ratios, a transformation was not necessary.

Elemental concentrations and stoichiometric ratios were compared between depth sections within each geomorphic feature as well as between geomorphic features within the same depth section. A one-way ANOVA was applied to determine differences between depth sections within each geomorphic feature. If differences were significant, the ANOVA was followed by a pairwise comparison test with tukey-correction. To determine differences between geomorphic features within each depth section, two-sample t-tests were applied. If homogeneity of variances was not given, the two-sample t-test was replaced by a Welch t-test.

For hypothesis (1a), differences of CUE, MBC, growth, respiration and turnover time were determined between the oxic palsa samples throughout all depth sections and the anoxic thermokarst samples throughout all depth sections. For this, a linear mixed effects model was applied, using the lmer function from the lm4 package (Bates et al., 2015). Here, the geomorphic feature was set as a fixed intercept while depth section was set as random intercept.

For hypothesis (1b), differences of CUE, MBC, growth, respiration and turnover time between geomorphic features were determined only for the same depth section and the same O_2 condition, meaning that only depth sections 4 and 5 were compared between palsa and

thermokarst. To determine if differences between geomorphic features were significant, two-sample t-tests were used to respectively compare all individual combinations within the same depth section and the same O_2 condition. If homogeneity of variances was still not given after data transformation, Welch t-tests were applied instead of the common t-tests. If normality of data was still not given after data transformation, the two-sample t-test was replaced by the Wilcoxon-Mann-Whitney test.

Hypothesis (2) tested differences of microbial parameters between depth sections of the oxic palsa samples as well as differences of microbial parameters between depth sections of the anoxic thermokarst. To assess if differences between depth sections within palsa or thermokarst were significant, a one-way ANOVA was applied, respectively, for each geomorphic feature. For MBC in the thermokarst as well as for respiration in the palsa, normality of residuals was not given even after data transformation. Here, the Kruskal-Wallis test was applied, followed by a dunn-test with bonferroni-correction from the “FSA” package (Ogle et al., 2025), if significant.

For hypothesis (3), differences between O_2 conditions were determined within the same geomorphic feature and the same depth section, meaning that again, only depth sections 4 and 5 were compared. Differences between O_2 condition were determined in the same way as differences between the geomorphic features by applying two-sample t-tests or, if not feasible, Welch’s t-test or the Wilcoxon-Mann-Whitney test.

Plots were created using the “ggplot2” package (Wickham, 2016). The boxplots contain the following elements: The thick horizontal line represents the median, the edges of the box represent the first and third quartile, the whiskers represent the minimum and maximum value, and dots represent outliers. Additionally, a principal component analysis (PCA) was performed for each geomorphic feature. For the palsa, the PCA included elemental concentrations and stoichiometric ratios as well as CUE, MBC, microbial growth, microbial respiration and turnover time of microbial biomass determined under oxic conditions. For the thermokarst, the PCA included elemental concentrations and stoichiometric ratios as well as CUE, MBC, microbial growth, microbial respiration and turnover time of microbial biomass determined under anoxic conditions. Correlations between all variables included the same data and were performed using the spearman correlation due to the lack of normal distribution. All values are given as mean ± standard deviation. Effects were treated as significant at $p < 0.05$.

3 Results

3.1 Elemental concentrations and stoichiometric ratios

Mean C concentrations varied between 457 ± 9.10 mg g peat^{-1} and 523 ± 6.52 mg g peat^{-1} while mean N concentrations ranged between 9.70 ± 4.20 mg g peat^{-1} and 24.1 ± 0.62 mg g peat^{-1} across the six depth sections and the two geomorphic features. The determined H concentrations were within the range of 59.9 ± 0.65 mg g peat^{-1} and 65.5 ± 2.62 mg g peat^{-1}, and mean C/N ratio varied between 25.4 ± 0.36 and 64.0 ± 28.6.

There were no significant differences in elemental concentrations or ratios between palsa and thermokarst, except for S content which was significantly higher in depth section 1 of the thermokarst than in depth section 1 of the palsa ($p < 0.05$; Fig. 2).

Depth affected elemental concentrations and ratios at both geomorphic features (Fig. 2). In the palsa, the H/C ratio was highest in the top depth sections and significantly decreased down to depth sections 4 and 5 ($p < 0.01$; Fig. 2g). Apart from that, elemental concentrations and ratios in the palsa were less strongly affected by depth than in the thermokarst. Nonetheless, there were trends observable in the palsa as well. N and S concentration tended to increase with depth while H and O concentration as well as C/N and O/C ratio showed a decreasing trend with depth (Fig. 2b, c, d, e, f, h). C concentration remained relatively stable throughout all depth sections with a mean value of 502 ± 20.8 mg g peat^{-1} (Fig. 2a).

In the thermokarst, all elemental concentrations and ratios were significantly affected by depth (Fig. 2). Depth section 0 of the thermokarst stood out by displaying significantly higher values for O concentration ($p < 0.01$), C/N ($p < 0.01$), H/C ($p < 0.0001$) and O/C ratio ($p < 0.001$) and significantly lower values for C and N concentration ($p < 0.01$ for both) when comparing with the other depth sections in the thermokarst, respectively. The lower depth sections showed comparably stable elemental concentrations, similar to what was observed in the palsa. However, in the thermokarst, there was a stronger variation in H concentrations than in the palsa (Fig. 2c). H concentration was highest in depth section 1 and lowest in depth sections 4 and 5 ($p < 0.01$). The H/C ratio showed relatively strong variations throughout all depth sections in the thermokarst, and the mean values ranged between 1.41 ± 0.01 and 1.63 ± 0.02. Here, the H/C ratio was highest in depth section 0 and significantly decreased from depth section 0 to 1 ($p < 0.001$), 1 to 2 ($p < 0.05$) and 2 to 4 ($p < 0.01$; Fig. 2h).

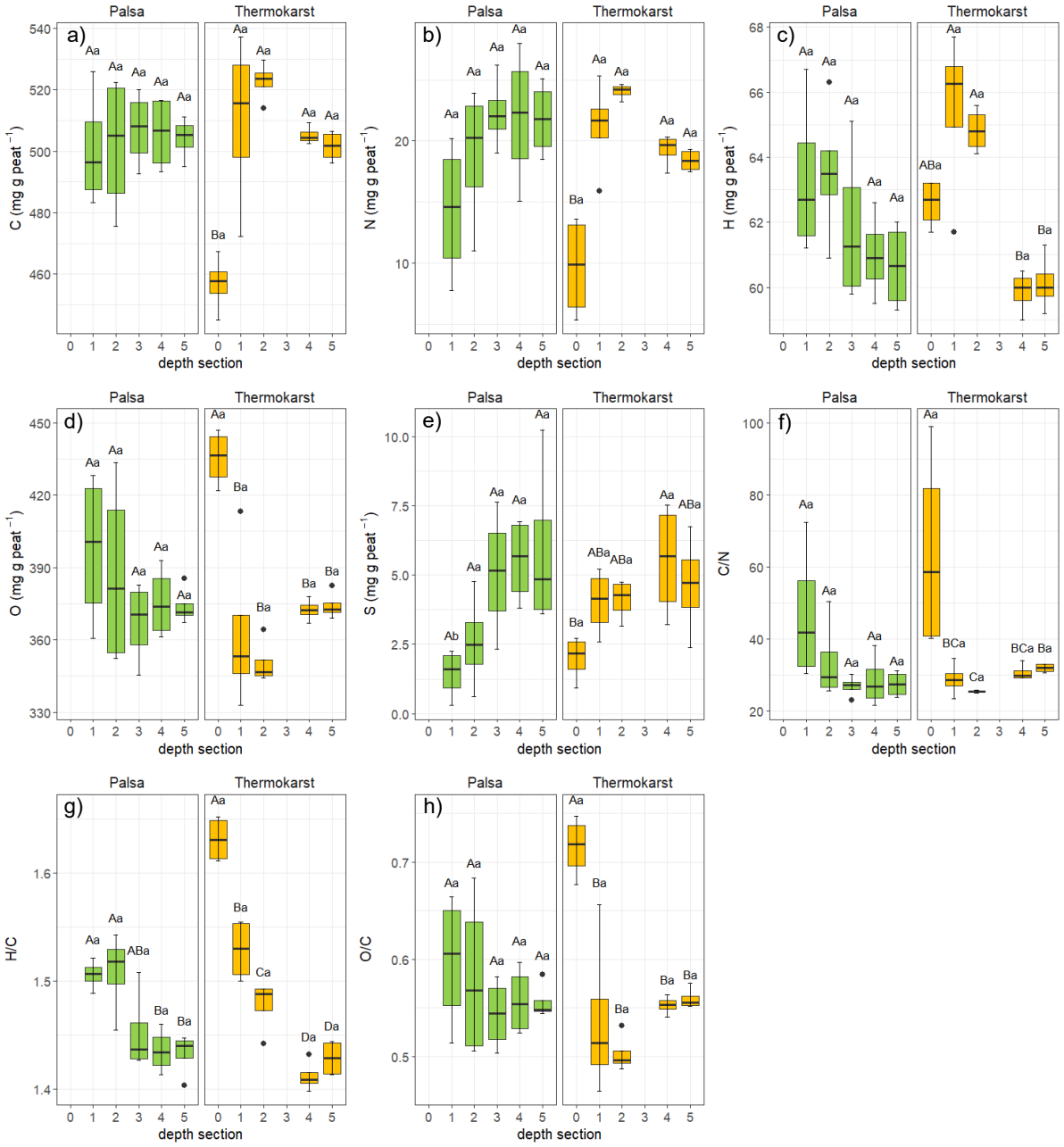


Figure 2: *Boxplots showing concentrations of a) carbon (C), b) nitrogen (N), c) hydrogen (H), d) oxygen (O) and e) sulfur (S) as well as of f) C/N, g) H/C and h) O/C ratio in different depth sections (0-5) of a Norwegian palsa and thermokarst. Different capital letters represent significant differences ($p < 0.05$) between depth sections within palsa or thermokarst, different small letters represent significant differences ($p < 0.05$) between palsa and thermokarst at the same depth section. Detailed statistics can be obtained from table A.3 and A.4.*

3.2 Microbial parameters in palsa vs. thermokarst

3.2.1 Different oxygen conditions

All microbial parameters except CUE and turnover time were significantly higher in the oxic palsa peat compared to the anoxic thermokarst peat when combining all depth sections (Tab.

2). The largest effect was found for microbial respiration. Here, the oxic palsa displayed a mean value of 44.0 ± 44.4 µg CO_2-C g peat^{-1} d^{-1} while mean respiration in the anoxic thermokarst was only 2.92 ± 1.51 µg CO_2-C g peat^{-1} d^{-1} ($p < 0.001$; Tab. 2). Turnover time was significantly lower in the oxic palsa peat than in the anoxic thermokarst peat ($p < 0.05$, Tab. 2).

Table 2: *Mean (± SD) microbial carbon use efficiency (CUE), microbial biomass carbon (MBC), microbial growth, microbial respiration and turnover time of microbial biomass in a Norwegian palsa (oxic) and thermokarst (anoxic) throughout all depth sections, as well as results from the statistical tests.*

Parameter	**Geomorphic feature**		**Statistics**	
	Palsa (oxic)	**Thermokarst** (anoxic)	**F-value**	**p-value**
CUE	0.37 ± 0.33	0.40 ± 0.35	0.03	0.88
MBC (µg C g peat^{-1})	525 ± 615	252 ± 329	7.78	**0.01**
Growth (µg C g peat^{-1} d^{-1})	61.8 ± 91.7	10.7 ± 25.3	13.4	**< 0.001**
Respiration (µg CO_2-C g peat^{-1} d^{-1})	44.0 ± 44.4	2.92 ± 1.51	229	**< 0.001**
Turnover time (d)	85.5 ± 130	178 ± 173	4.75	**0.04**

3.2.2 Same oxygen conditions

Oxic

Microbial CUE was significantly higher in the palsa in depth section 5 with a mean of 0.27 ± 0.23 compared to the thermokarst with a mean of 0.02 ± 0.02 ($p < 0.05$; Tab. 3). The MBC displayed no significant difference between palsa and thermokarst when comparing the same depth sections under oxic conditions, but it showed overall higher values in the palsa compared to the thermokarst (Tab. 3, Fig. 3b). Microbial growth was significantly higher in the palsa compared to the thermokarst in depth section 5 ($p < 0.05$; Tab. 3). Similarly, there was higher microbial growth in the palsa than in the thermokarst in depth section 4, but this difference was not significant (Tab. 3, Fig. 3c). Microbial respiration did not display significant differences between palsa and thermokarst, and the variability of values in the palsa was high (Tab. 3, Fig. 3d). Microbial turnover time was significantly higher in the thermokarst than in the palsa in depth section 5 where it displayed a mean value of 150 ± 74.0 days in the thermokarst and a mean value of 35.9 ± 35.3 days in the palsa ($p < 0.05$; Tab. 3).

Anoxic

Under anoxic conditions, microbial CUE was significantly higher in the palsa compared to the thermokarst in depth section 4 ($p < 0.05$; Tab. 3). Although not significant, CUE in depth section

5 under anoxic conditions also showed higher values in the palsa, with a mean of 0.48 ± 0.37, than in the thermokarst, with a mean of 0.18 ± 0.21 (Tab. 3, Fig. 3a). For both MBC and microbial growth, differences between palsa and thermokarst were not significant (Tab. 3). Microbial respiration was significantly lower in the palsa compared to the thermokarst in depth section 5 ($p < 0.05$), and a similar tendency was also observable for depth section 4 (Tab. 3, Fig. 3d). Microbial turnover under anoxic conditions did not vary significantly between the two geomorphic features (Tab. 3).

Table 3: *Mean (± SD) microbial carbon use efficiency (CUE), microbial biomass carbon (MBC), microbial growth, microbial respiration and turnover time of microbial biomass in a Norwegian palsa and thermokarst at the same oxygen condition (oxic, anoxic) and in the same depth section (4, 5), as well as results from the statistical tests. The t-value is the test statistic derived from the two-sample t-test or the Welch t-test. The W-value is the test statistic derived from the Wilcoxon-Mann-Whitney-test (values in italics).*

Parameter	**Oxygen condition & depth section**	**Geomorphic feature**		**Statistics**	
		Palsa	**Thermokarst**	**t/ *W***	**p-value**
CUE	Oxic 4	0.09 ± 0.07	0.06 ± 0.06	0.78	0.47
	Oxic 5	0.27 ± 0.23	0.02 ± 0.02	*16.0*	**0.03**
	Anoxic 4	0.64 ± 0.35	0.16 ± 0.08	2.96	**0.03**
	Anoxic 5	0.48 ± 0.37	0.18 ± 0.21	1.38	0.23
MBC	Oxic 4	325 ± 284	170 ± 145	0.85	0.43
(µg C g peat^{-1})	Oxic 5	486 ± 590	78.6 ± 57.2	*15.0*	0.05
	Anoxic 4	153 ± 135	80.0 ± 60.0	*13.0*	0.18
	Anoxic 5	148 ± 113	83.6 ± 40.5	0.89	0.42
Growth	Oxic 4	8.64 ± 12.1	2.06 ± 1.81	0.92	0.39
(µg C g peat^{-1} d^{-1})	Oxic 5	63.8 ± 107	0.88 ± 1.16	2.80	**0.03**
	Anoxic 4	10.3 ± 12.6	0.47 ± 0.24	2.29	0.09
	Anoxic 5	3.75 ± 5.16	0.95 ± 1.31	0.84	0.44
Respiration	Oxic 4	56.9 ± 49.9	32.8 ± 3.68	0.85	0.43
(µg CO_2-C g peat^{-1}	Oxic 5	76.9 ± 82.2	35.0 ± 7.87	*12.0*	0.31
d^{-1})	Anoxic 4	1.62 ± 0.50	2.64 ± 0.81	*1.00*	0.05
	Anoxic 5	1.68 ± 0.39	4.20 ± 2.22	-3.21	**0.02**
Turnover time	Oxic 4	74.0 ± 51.6	104 ± 50.2	-0.88	0.41
(d)	Oxic 5	35.9 ± 35.3	150 ± 74.0	-2.93	**0.03**
	Anoxic 4	83.6 ± 92.2	187 ± 109	-1.55	0.17
	Anoxic 5	159 ± 129	250 ± 220	-0.54	0.61

3.3 Depth gradients of microbial parameters

3.3.1 Palsa

Depth did not significantly affect the CUE in the palsa under oxic conditions, but trends were visible (Tab. 4, Fig. 3a). In the top two depth sections, medium to high values were observed, ranging between 0.50 ± 0.38 and 0.69 ± 0.23 (Tab. 4). For depth sections 3 to 5 rather low values were determined, ranging between 0.09 ± 0.07 and 0.28 ± 0.41 (Tab. 4).

Mean MBC in the palsa varied between 227 ± 218 µg C g peat^{-1} and 1,254 ± 935 µg C g peat^{-1} (Tab. 4). MBC did not show significant differences along the profile (Tab. 4). However, MBC was highest in the top depth section and lowest in depth section 2 (Tab. 4, Fig. 3b). Across depth sections 2 to 5, MBC remained relatively constant, especially when focusing on the median (Fig. 3b).

Similarly, microbial growth did not significantly vary with increasing depth in the palsa, but it reached its maximum value in the top depth section (Tab. 4). The data distribution in the box-plot showed a comparable growth rate in depth sections 1 and 2 and then lower but relatively stable growth rates in depth sections 3 to 5 (Fig. 3c). Mean microbial growth showed a slightly different pattern than the medians shown in the boxplots with similar values in depth sections 2, 3 and 5 and a minimum reached in depth section 4 (Tab. 4).

Table 4: *Mean (± SD) microbial carbon use efficiency (CUE), microbial biomass carbon (MBC), microbial growth, microbial respiration and turnover time of microbial biomass in depth sections 1 to 5 of a Norwegian palsa under oxic conditions, as well as results from the statistical tests. The F-value is the test statistic derived from the one-way ANOVA. The X^2-value is the test statistic derived from the Kruskal Wallis Test (values in italics).*

Parameter	**Depth section**					**Statistics**	
	1	**2**	**3**	**4**	**5**	**F/ X^2**	**p-value**
CUE	0.50 ± 0.38	0.69 ± 0.23	0.28 ± 0.41	0.09 ± 0.07	0.27 ± 0.23	2.85	0.06
MBC (µg C g peat^{-1})	1,254 ± 935	227 ± 218	333 ± 330	325 ± 284	486 ± 590	2.36	0.10
Growth (µg C g peat^{-1} d^{-1})	100 ± 106	61.8 ± 59.6	74.1 ± 144	8.64 ± 12.1	63.8 ± 107	1.26	0.33
Respiration (µg CO_2-C g peat^{-1} d^{-1})	42.1 ± 10.8	15.7 ± 4.22	28.4 ± 6.70	56.9 ± 49.9	76.9 ± 82.2	*11.7*	**0.02**
Turnover time (d)	123 ± 157	5.18 ± 4.47	189 ± 222	74.0 ± 51.6	35.9 ± 35.3	1.22	0.34

The determined microbial respiration was within the range of 15.7 ± 4.22 µg CO_2-C g $peat^{-1}$ d^{-1} and 76.9 ± 82.2 µg CO_2-C g $peat^{-1}$ d^{-1} (Tab. 4). Microbial respiration varied significantly with depth in the palsa under oxic conditions ($p < 0.05$; Tab. 3). Respiration decreased from depth section 1 to 2 ($p < 0.05$) and then increased up to depth section 5 where the maximum was reached (Tab. 4, Fig. 3d). In consequence, respiration in depth section 2 was significantly lower than in depth section 5 ($p < 0.05$).

The turnover time of microbial biomass displayed a high variability, and there was no significant difference between depth sections (Tab. 4, Fig. 3e).

3.3.2 Thermokarst

Under anoxic conditions, highest and lowest CUE in the thermokarst occurred at depth section 2 and 4, respectively, however, depth had no significant effect on CUE (Tab. 5). CUE in the thermokarst showed a similar pattern with depth as observed for the palsa. In the top three depth sections CUE ranged between 0.41 ± 0.04 and 0.72 ± 0.39 (Tab. 5). In depth sections 4 and 5 rather low values were found, ranging between 0.16 ± 0.08 and 0.18 ± 0.21 (Tab. 5).

Table 5: *Mean (± SD) microbial carbon use efficiency (CUE), microbial biomass carbon (MBC), microbial growth, microbial respiration and turnover time of microbial biomass in depth sections 0, 1, 2, 4 and 5 of a Norwegian thermokarst under anoxic conditions, as well as results from the statistical tests. The F-value is the test statistic derived from the one-way ANOVA. The X^2-value is the test statistic derived from the Kruskal Wallis Test (values in italics).*

Parameter	**Depth section**					**Statistics**	
	0	**1**	**2**	**4**	**5**	**F/ X^2**	**p-value**
CUE	0.41 ± 0.04	0.56 ± 0.45	0.72 ± 0.39	0.16 ± 0.08	0.18 ± 0.21	2.22	0.12
MBC (µg C g $peat^{-1}$)	689 ± 290	122 ± 133	286 ± 472	80.0 ± 60.1	83.6 ± 40.5	*8.82*	0.07
Growth (µg C g $peat^{-1}$ d^{-1})	2.52 ± 1.14	29.0 ± 51.4	18.5 ± 17.9	0.47 ± 0.24	0.95 ± 1.31	1.79	0.19
Respiration (µg CO_2-C g $peat^{-1}$ d^{-1})	3.78 ± 1.10	1.98 ± 1.10	1.98 ± 0.89	2.64 ± 0.81	4.20 ± 2.22	2.64	0.08
Turnover time (d)	253 ± 111	157 ± 270	60.7 ± 78.4	187 ± 109	250 ± 220	1.32	0.31

Depth section 0 displayed a higher mean MBC with a value of 689 ± 290 µg C g peat^{-1} than depth sections 1 to 5, but this difference was in no case significant (Tab. 5). The MBC values of depth sections 1 to 5 were relatively constant throughout the profile, especially when focusing on the data distribution shown in the boxplot (Fig. 3b).

In depth section 0 microbial growth was lower than in depth sections 1 and 2 and higher than in depth sections 4 and 5 (Tab. 5, Fig. 3c). However, no significant differences between depth sections were found for microbial growth in the thermokarst (Tab. 5).

Microbial respiration in the thermokarst varied between 1.98 ± 0.89 µg CO_2-C g peat^{-1} d^{-1} and 4.20 ± 2.22 µg CO_2-C g peat^{-1} d^{-1} (Tab. 5). In contrast to the palsa, values in the thermokarst varied less strongly with depth, and no significant differences were found (Tab. 5, Fig. 3d).

Microbial turnover in the thermokarst varied strongly within the single depth sections, and no significant differences were found between the depth sections (Tab. 5, Fig. 3e).

3.4 Microbial parameters under oxic vs. anoxic conditions

Palsa

CUE was almost seven times higher under anoxic conditions than under oxic conditions in depth section 4 ($p < 0.05$; Tab. 6). In contrast, MBC did not vary significantly with O_2 condition as variability of the replicates was high (Tab. 6, Fig. 3b). Similarly, there was no significant difference observable when comparing microbial growth between oxic and anoxic conditions (Tab. 6). Mean growth rates were both higher and lower under oxic compared to anoxic conditions, depending on depth section, and variability was high (Tab. 6, Fig. 3c). Microbial respiration rates were significantly higher under oxic conditions compared to anoxic conditions in depth section 4 as well as in depth section 5 ($p < 0.001$ for both). In depth section 5, respiration was more than 45 times higher under oxic conditions than under anoxic conditions. CH_4 production was barely elevated under both oxic and anoxic conditions (Tab. A.5, A.6). Microbial turnover time did not vary between oxic and anoxic conditions (Tab. 6).

Thermokarst

CUE in the thermokarst displayed higher but not significantly higher values under anoxic compared to oxic conditions (Tab. 6, Fig. 3a). Comparably, MBC and microbial growth did not significantly vary between O_2 conditions (Tab. 6, Fig. 3b, c). Similar to the palsa, microbial respiration was significantly higher under oxic conditions compared to anoxic conditions in depth section 4 and depth section 5 ($p < 0.001$ for both; Tab. 6). The smallest difference between oxic and anoxic conditions was found for depth section 4 (Tab. 5). Here, mean microbial respiration under oxic conditions was 32.8 ± 3.68 µg CO_2-C g peat^{-1} d^{-1} while mean microbial

respiration under anoxic conditions was 2.64 ± 0.81 µg CO_2-C g peat^{-1} d^{-1} (Tab. 5). Hence, microbial respiration was at least 12 times higher under oxic conditions than under anoxic conditions. No or very little elevated CH_4 production was measured under oxic conditions after 48 hour incubation (Tab. A.5). Comparably, no CH_4 production was detected under anoxic conditions at the end of the incubation period of 160 hours (Tab. A.6). Even after 91 day incubation under water saturated and anoxic conditions, CH_4 production remained very low (Tab. A.7). Furthermore, there were no significant differences in microbial turnover time between oxic and anoxic conditions (Tab. 6).

Table 6: *Mean (± SD) microbial carbon use efficiency (CUE), microbial biomass carbon (MBC), microbial growth, microbial respiration and turnover time of microbial biomass under oxic and anoxic conditions at the same geomorphic feature (palsa, thermokarst) and in the same depth section (4, 5), as well as results from the statistical tests. The t-value is the test statistic derived from the two-sample t-test or the Welch t-test. The W-value is the test statistic derived from the Wilcoxon-Mann-Whitney test (values in italics).*

Parameter	**Geomorphic feature & depth section**	**Oxygen condition**		**Statistics**	
		Oxic	**Anoxic**	**t/ *W***	**p-value**
CUE	Palsa 4	0.09 ± 0.07	0.64 ± 0.35	-3.64	**0.01**
	Palsa 5	0.27 ± 0.23	0.48 ± 0.37	-0.86	0.43
	Thermokarst 4	0.06 ± 0.06	0.16 ± 0.08	-2.06	0.09
	Thermokarst 5	0.02 ± 0.02	0.18 ± 0.21	*2.50*	0.12
MBC	Palsa 4	325 ± 284	153 ± 135	1.00	0.36
(µg C g peat^{-1})	Palsa 5	486 ± 590	148 ± 113	*10.0*	0.66
	Thermokarst 4	170 ± 145	80.0 ± 60.0	*13.0*	0.18
	Thermokarst 5	78.6 ± 57.2	83.6 ± 40.5	*7.00*	0.87
Growth	Palsa 4	8.64 ± 12.1	10.3 ± 12.6	-0.09	0.93
(µg C g peat^{-1} d^{-1})	Palsa 5	63.8 ± 107	3.75 ± 5.16	1.51	0.19
	Thermokarst 4	2.06 ± 1.81	0.47 ± 0.24	2.24	0.07
	Thermokarst 5	0.88 ± 1.16	0.94 ± 1.31	0.11	0.92
Respiration	Palsa 4	56.9 ± 49.9	1.62 ± 0.50	8.23	**< 0.001**
(µg CO_2-C g peat^{-1}	Palsa 5	76.9 ± 82.2	1.68 ± 0.39	7.83	**< 0.001**
d^{-1})	Thermokarst 4	32.8 ± 3.68	2.64 ± 0.81	17.1	**< 0.001**
	Thermokarst 5	35.0 ± 7.87	4.20 ± 2.22	8.53	**< 0.001**
Turnover time	Palsa 4	74.0 ± 51.6	83.6 ± 92.2	0.20	0.85
(d)	Palsa 5	35.9 ± 35.3	159 ± 129	-1.84	0.13
	Thermokarst 4	104 ± 50.2	187 ± 109	-1.31	0.24
	Thermokarst 5	150 ± 74.0	250 ± 220	-0.63	0.56

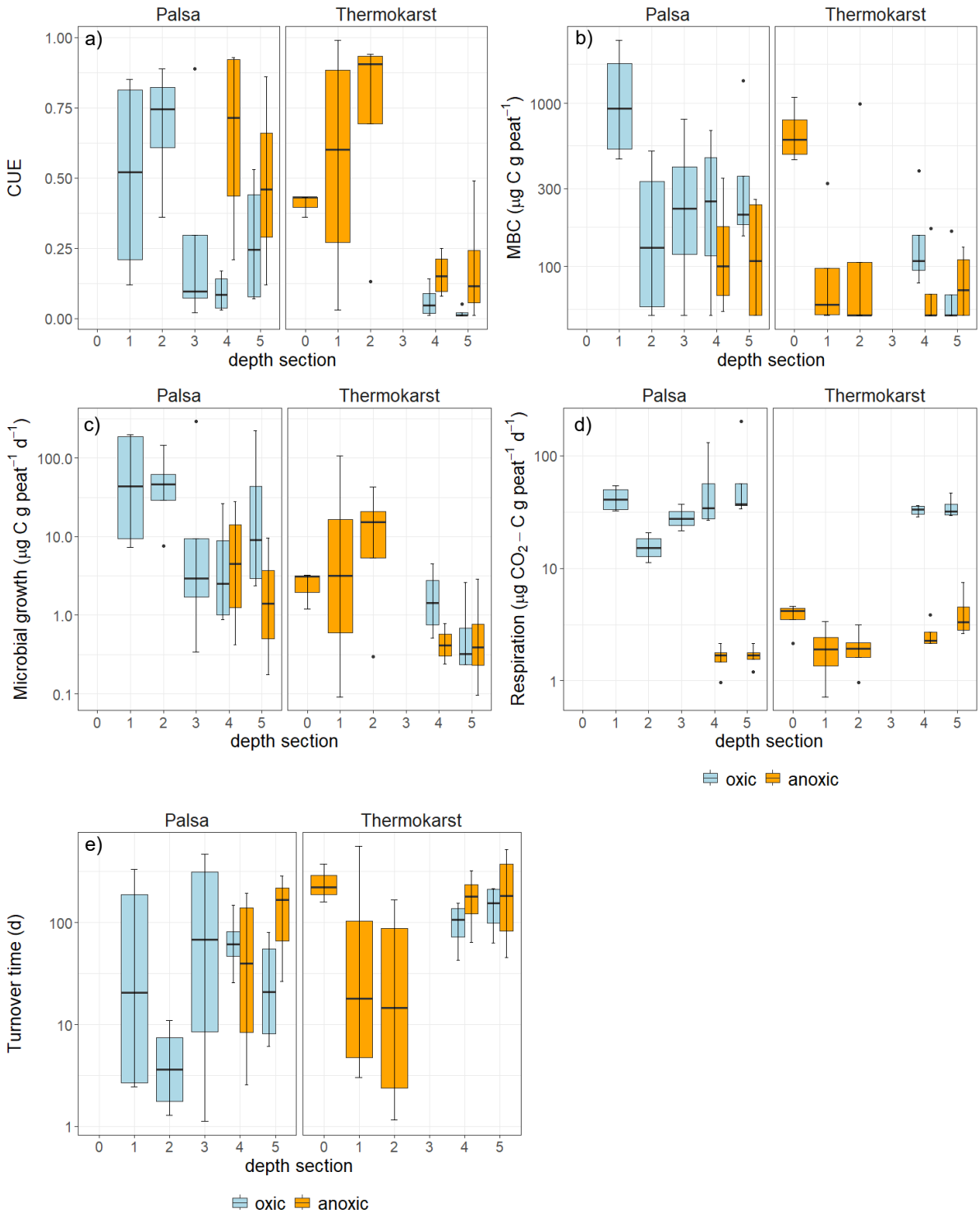


Figure 3: *Boxplots showing a) microbial carbon use efficiency (CUE), b) microbial biomass carbon (MBC), c) microbial growth, d) microbial respiration and e) turnover time of microbial biomass in a Norwegian palsa and thermokarst in different depth sections (0-5) and at different oxygen conditions (oxic, anoxic).*

3.5 Influence of elemental concentrations and stoichiometric ratios on microbial parameters

The first two dimensions of the PCA from the palsa together explained 72.9% of the variability within the data ($p < 0.01$; Fig. 4a). Elemental concentrations, ratios and CUE contributed most to the first two dimensions of the PCA (Fig. A.1a). Microbial respiration contributed the least, as also indicated by a small arrow size (Fig. 4a, Fig. A.1a). The strongest correlations were found between CUE, growth and H concentration which were positively correlated ($p < 0.01$, CUE and growth: $p < 0.0001$; Fig. 4a). Additionally, H/C ratio showed a positive correlation with CUE ($p < 0.01$) and microbial growth ($p < 0.05$; Fig. 4a). O concentration was positively correlated with MBC ($p < 0.05$; Fig. 4a). Depth sections varied most strongly along PC2 (Fig. 4a). Depth sections 1 and 2 were found mostly in the positive range of PC2 and appeared to mostly explain CUE, growth, H and H/C. Depth sections 3 to 5 were mostly present in the negative range of PC2.

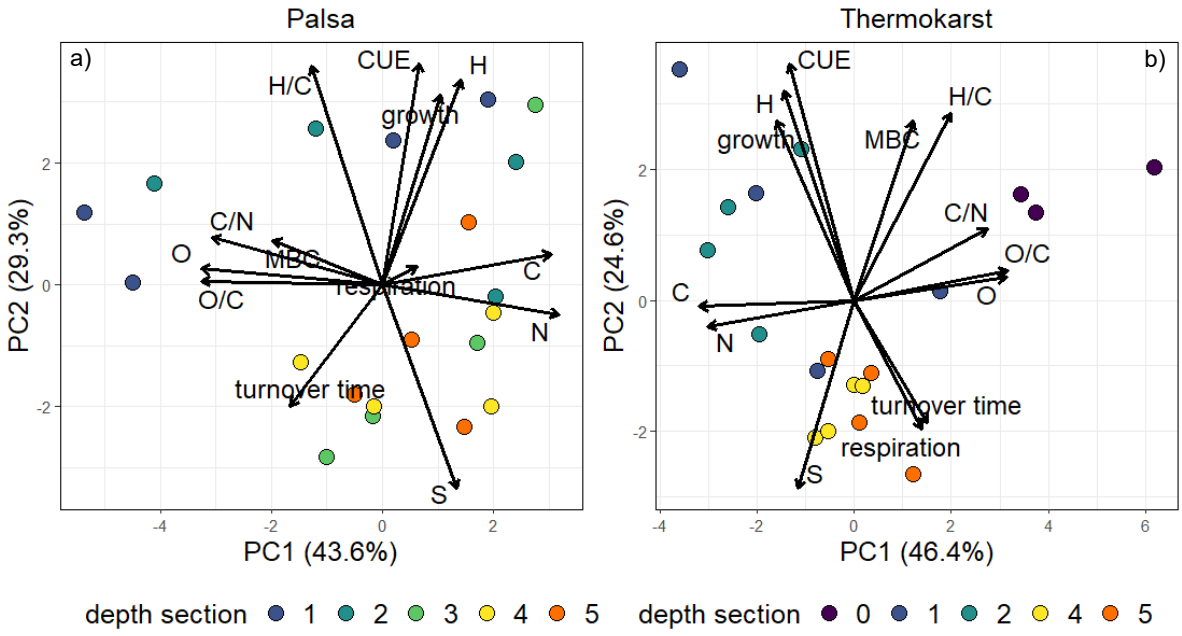


Figure 4: *Principal component analysis (PCA) of microbial parameters (carbon use efficiency (CUE), microbial biomass carbon (MBC), microbial growth, microbial respiration and turnover time of microbial biomass) as well as elemental concentrations and stoichiometric ratios a) in a Norwegian palsa and b) in a Norwegian thermokarst. Dots represent the individual samples in different depth sections (0-5), arrows represent the variables. Spearman correlation coefficients can be obtained from table A.8 and A.9.*

The first two dimensions of the PCA from the thermokarst significantly explained the observed variability within the data ($p < 0.01$). Together, PC1 and PC2 explained 71.0% of the PCA (Fig.

4b). The PCA was most strongly influenced by elemental concentrations and ratios as well as CUE and least by respiration (Fig. A.1b). CUE and growth were significantly (CUE, $p < 0.05$) or strongly but not significantly (growth) correlated with H concentration (Fig. 4b). Additionally, MBC and H/C were positively correlated ($p < 0.05$; Fig. 4b). Similar to the palsa, depth sections varied along PC2 as samples from depth sections 0 to 2 were found mostly in the positive range of PC2 while samples from depth sections 4 and 5 were present in the negative range of PC2 (Fig. 4b). Additionally, a variation along PC1 was observable when focusing on the top depth sections. Depth sections 1 and 2 were mostly present in the negative range of PC1, clustering close to CUE, growth, H, C and N. Samples from depth section 0 occurred in the positive range of PC1 (Fig. 4b).

4 Discussion

4.1 Literature comparison of microbial parameters

Overall, the determined CUEs in this study are in line with literature. The mean CUE varied between 0.02 and 0.72 across all depth sections and the two permafrost features. Mean CUEs reported throughout various publications on peat or arctic soils are within the range of 0.03 and 0.70 (Hill et al., 2014; Sihi et al., 2017; Chen et al., 2018; Lee et al., 2019; Soares & Rousk, 2019). However, the reported CUEs varied less strongly than the values observed here and often times did not reach values way higher than 0.30 (Hill et al., 2014; Lee et al., 2019; Soares & Rousk, 2019). Hence, the determined values in this study point to a relatively high microbial CUE in the top peat sections. These comparably high CUE values might have been caused by differences in the applied methods as none of the mentioned publications used the ^{18}O-method to determine the CUE. For instance, Hill et al. (2014) calculated the CUE in peat based on N, phosphorous (P) and C availability. Chen et al. (2018) similarly assessed CUE of a permafrost swamp meadow based on C/N stoichiometry, and Lee et al. (2019) performed a simulation of CUE based on the ratio of net primary production to gross primary production for different Norwegian permafrost soils. Moreover, methodological limitations, especially regarding the ^{18}O enrichment and the determination of MBC, might explain the large variation of CUE in this study (see chapter 4.5).

The values for MBC were generally in the range reported for other peatlands (Tab. 7). For instance, Song et al. (2014) determined a mean MBC of 606 µg C g peat^{-1} under oxic conditions and of 499 µg C g peat^{-1} under anoxic conditions in a Chinese permafrost peatland which is similar to the 525 µg C g peat^{-1} determined in this study under oxic conditions for the palsa. However, it is above the 252 µg C g peat^{-1} determined for the anoxic thermokarst. Moreover, two studies found much higher MBC in peat than that documented for palsa and thermokarst in this study (Keiser et al., 2019; Chen et al., 2022). The reason for the higher values determined by Chen et al. (2022) is unclear. The higher values determined by Keiser et al. (2019) for a boreal peat bog might stem from a performed addition of different substrates to their samples. Overall, the literature comparison points to a rather low microbial biomass in Áidejávri, especially in the thermokarst, which also suggests low growth and respiration rates.

Microbial activity in peat from Áidejávri was, in fact, rather low compared to other studies. Bacterial growth reported by Soares and Rousk (2019) for subarctic peat in Sweden that was incubated at 22 °C was similar or higher than the microbial growth reported for the palsa in this study and higher than that of the thermokarst. This might have been the case because Soares and Rousk (2019) determined growth with the use of methyl-^{3}H thymidine instead of the ^{18}O-method that was applied here. Moreover, the peat in the study of Soares and Rousk (2019)

was not affected by permafrost. Similarly, mean respiration rates documented here tended to be lower than those reported from other studies for permafrost peatlands (Tab. 8). However, a comparison of values is limited because conditions of laboratory incubations and sample treatment can differ (Turetsky, 2004; Kjær et al., 2024). For instance, Kirkwood et al. (2021) incubated their anoxic samples for a longer time period. Moreover, incubation temperatures between the mentioned publications vary between 4 °C and 20 °C (Tab. 8). The low MBC, growth and respiration values suggest that microbes contribute little to the organic C pool of the studied permafrost peatlands as most of the peat C pool consists of hardly undecomposed plant litter (Sjörs, 1980; Rydin & Jeglum, 2013).

Mean turnover times of microbial biomass were mostly in the range that was reported in literature for mineral, non-permafrost soils. Here, mean turnover times reported from various studies range between 18 and 322 days (Sakamoto & Hodono, 2000; Cheng, 2009; Spohn et al., 2016a; Spohn et al., 2016b; Pei et al., 2021; Tian et al., 2023). This shows that in the thawed palsa peat under oxic conditions, the mean turnover time was similar to that reported for mineral, non-permafrost soils. The mean turnover time of the thermokarst can be classified more in the upper range of the values reported from other publications. This indicates that the turnover of microbial biomass might take longer in water-logged, anoxic thermokarsts of the permafrost zone compared to oxic, non-permafrost soils with higher in-situ temperatures.

Table 7: *Mean (± SD) microbial biomass carbon (MBC) in peat of permafrost and peatland ecosystems at different oxygen conditions and different incubation temperatures.*

Ecosystem	**Location**	**Oxygen condition**	**Incubation temperature**	**MBC** (µg C g peat^{-1})	**Reference**
Palsa	Norway	oxic	15 °C	525 ± 615	This study
Permafrost peatland	China	oxic	10 °C	606 ± 390	Song et al., 2014
Permafrost peatland	China	anoxic	10 °C	499 ± 106	Song et al., 2014
Open permafrost peatland	China	oxic	n.a.	10,770 ± 5,470	Chen et al., 2022
Alpine meadow	China	oxic	4 °C	202 ± 85.9	Dong et al., 2023
Thermokarst	Norway	anoxic	15 °C	252 ± 329	This study
Boreal peat bog	USA	oxic	14 °C, 20°C and 28 °C	4,760 ± 1,050	Keiser et al., 2019

Alpine swamp meadow	China	oxic	4 °C	443 ± 231	Shang et al., 2015
Peatland, growing season	China	oxic	25 °C	538 ± 85.9	Luo et al., 2022

Table 8: *Mean (± SD) microbial respiration in peat of permafrost and peatland ecosystems at different oxygen conditions and different incubation temperatures. Values without standard deviation were depicted from figures of the published study.*

Ecosystem	**Location**	**Oxygen condition**	**Incubation temperature**	**Respiration** (µg CO_2-C g peat^{-1} d^{-1})	**Reference**
Palsa	Norway	oxic	15 °C	44.0 ± 44.4	This study
Palsa	Norway	oxic	10 °C	92.6 ± 40.1	Kjær et al., 2024
Palsa	Canada	anoxic	4 °C and 14 °C	8.00	Kirkwood et al., 2021
Permafrost peatlands	Alaska	oxic	20 °C	130	Treat et al., 2014
Permafrost peatlands	Alaska	anoxic	20 °C	66.0	Treat et al., 2014
Thermokarst	Norway	anoxic	15 °C	2.92 ± 1.51	This study
Thermokarst	Canada	anoxic	4 °C and 14 °C	10.0	Kirkwood et al., 2021
Thermokarst	Norway	oxic	10 °C	115 ± 117	Kjær et al., 2024

4.2 Microbial parameters in palsa vs. thermokarst

4.2.1 Different oxygen conditions

Hypothesis (1a) tested if microbial parameters differed between oxic palsa peat and anoxic thermokarst peat. As palsas usually display oxic conditions (Lawrence et al., 2015) while thermokarsts present water-logged, anoxic conditions (Jorgenson & Osterkamp, 2005; Connon et al., 2014; Zandt et al., 2020), this comparison represents the in-situ conditions of both geomorphic features best and could give relevant insights into microbial C allocation at the site.

Hypothesis (1a) was not confirmed for CUE as no significant differences were found between the two geomorphic features. CUE was very similar at both geomorphic features with a value of 0.37 in the oxic palsa and 0.40 in the anoxic thermokarst. The prevailing oxic conditions (Lawrence et al., 2015) and an anticipated higher N availability and lower C/N ratio (Pengerud et al., 2013; Chen et al., 2018; Kirkwood et al., 2021) as well as a higher degradability of OM (Hodgkins et al., 2014) suggested a higher CUE in the palsa (Manzoni et al., 2012). However, it seems that nutrient contents and stoichiometric ratios had no effect on CUE and did, in fact, barely vary between palsa and thermokarst (see chapter 4.2.2). Moreover, CUE of both geomorphic features varied similarly along the depth gradients (see chapter 4.3), and anoxic conditions possibly had no or an increasing effect on CUE (see chapter 4.4). This could explain why CUE did not differ between oxic palsa and anoxic thermokarst. The results show that microorganisms in palsa and thermokarst allocate proportionally similar amounts of C into growth, despite changes of the microbial community with thermokarst as indicated by differences in overall microbial activity and biomass.

Microbial biomass, growth and respiration were significantly higher in the oxic palsa compared to the anoxic thermokarst. The differences were likely not caused by peat characteristics, as no differences were found here (see chapter 4.2.2), but rather by the contrasting O_2 conditions. This was expected and is in line with studies that performed in-situ respiration measurements at palsas and thermokarsts (Bäckstrand et al., 2010; Estop-Aragonés et al., 2018b). The results indicate that palsas store more microbial C in total compared to thermokarsts. At the same time, more C in total might be lost to respiration – and in that way fuel global warming – from well oxidated palsas than from water saturated, anoxic thermokarsts. Similar conclusions were drawn by others (Schädel et al., 2016; Estop-Aragonés et al., 2018a; Estop-Aragonés et al., 2018b). The overall lower microbial activity in the thermokarst is also in line with the significantly higher microbial turnover times documented for the thermokarst. Thus, in-situ conditions of thermokarsts appear to restrict microbial activity stronger compared to those of palsas.

4.2.2 Same oxygen conditions

Hypothesis (1b) compared microbial parameters between permafrost peat from the palsa and former permafrost peat from the thermokarst at the same depth section and under the same O_2 condition. This comparison could help to disclose if and how microbial parameters differ between palsa and thermokarst when the variables depth and O_2 availability are constant.

Oxic

Under oxic conditions, CUE tended to be higher in the palsa than in the thermokarst but was only significantly different in depth section 5. This observation partially agrees with the

hypothesis (1b). In agreement with this result, Chen et al. (2018) observed a decreasing CUE with thermokarst under oxic conditions in a 16 year-old collapsed permafrost meadow. These results indicate that in thermokarst peat more C has to be respired in order to achieve a similar growth to that of palsa peat when O_2 conditions of both geomorphic features are oxic.

It was expected that CUE in the thermokarst would be lower than in the palsa because lower N availability and a higher C/N ratio were anticipated for the thermokarst (Pengerud et al., 2013; Chen et al., 2018; Kirkwood et al., 2021), and this can limit CUE (Ågren et al., 2001; Manzoni & Porporato, 2009; Manzoni et al., 2012; Sinsabaugh et al., 2013; Sinsabaugh et al., 2016; Chen et al., 2018; Fang et al., 2016). In this study, there were no differences in C/N ratio between deep palsa peat and deep thermokarst peat. Moreover, C/N ratio was not found to be correlated with CUE in palsa or thermokarst. It is possible that the variability of C/N in this study was too low to observe effects on the CUE and that an overall low N availability restricted a significant change in CUE with changing C/N ratio. Comparably, there were no significant differences in N concentration between palsa and thermokarst, and no relationship between N concentration and CUE was observed. In consequence, N concentration and C/N ratio do not explain the higher CUE observed in the palsa peat.

It was further expected that degradability of OM would affect the CUE at the studied geomorphic features (Blagodatskaya & Kuzyakov, 2008; Manzoni et al., 2012; Öquist et al., 2017). However, the OM can be characterized as recalcitrant in both the permafrost palsa peat and the corresponding thermokarst peat. This assumption is based on D'Andrilli et al. (2015) who propose for marine and glacial ecosystems that an H/C ratio ≥ 1.5 indicates elevated proportion of degradable OM while an H/C ratio < 1.5 indicates more recalcitrant OM. Labile organic materials, such as cellulose, consist of more H relative to C than recalcitrant materials, such as lignin (Grannas et al., 2006; Moore et al. 2018). Therefore, a high H/C ratio is indicative for the presence of more easily degradable materials and a less advanced decomposition (D'Andrilli et al., 2015; Moore et al., 2018; Zaccone et al., 2018; Abdelrahman et al., 2023). Additionally, the humification index (based on the ratio of wavenumbers 1630/1090) that was derived for the same site (Henrich, 2025) shows a similar pattern as the H/C ratio here, with an increasing value when H/C ratio is decreasing. This supports the assumption that, in this study, a higher H/C ratio is indicative for a larger presence of degradable compounds and, hence, could point to a higher CUE. In fact, H/C ratio and CUE were positively correlated, but the correlation was only significant in the palsa. Additionally, H content was significantly correlated with the CUE at both geomorphic features and displayed the strongest correlation in the palsa. This suggests that, due to overall high concentration and low variability of C, a higher H content alone could point to an overall lower degree of dehydrogenation of the OM, indicating a higher content of undecomposed substrates (D'Andrilli et al., 2015; Moore et al., 2018; Abdelrahman et al.,

2023). Thus, the degradability of OM seems to control CUE at the studied site stronger than N concentration or C/N ratio, similar to what was observed by Takriti et al. (2018).

As H/C ratio and H content did not differ between palsa and thermokarst, it appears like OM decomposition did not strongly advance in the deep thermokarst peat compared to the permafrost palsa peat, despite it being thawed and possibly exposed to higher decomposition for decades. The overall similar elemental concentrations and ratios in palsa and thermokarst, including C/N, additionally indicate that thermokarst did not result in a considerable change of peat composition, at least not yet. This is a result of the previous oxic conditions in the former palsa (before collapse to thermokarst), which limit microbial activity in the present anoxic thermokarst (see chapter 4.2.1). Moreover, the already low amount of decomposable C in the permafrost layer of the palsa studied here might have hindered further decomposition after permafrost collapse. Other studies reported a high degradability of permafrost peat due to a preservation of the peat when frozen (Zimov et al., 2006; Waldrop et al., 2010). The results of this study indicate that the permafrost peat in Áidejávri does not store a high proportion of degradable substrates. Heffernan et al. (2020) and Estop-Aragonés et al. (2022) explain similar observations with a processing of the OM before incorporation into permafrost. In fact, the palsa in Áidejávri developed epigenetically (Henrich, 2025), and the peat was possibly already exposed to decomposition before permafrost developed. The results further suggest that no substantial amounts of easily degradable substrates have been accumulated under anoxic conditions during thermokarst development. The overall H/C ratios and H contents documented here are in line or slightly higher than values reported for peat in other studies (Zaccone et al., 2007; Fernandes et al., 2010; Reiche et al., 2010; Moore et al., 2018; Turunen et al., 2024). Harris et al. (2023) and Heffernan et al. (2020) similarly did not observe significant differences in the degree of peat decomposition after thaw of permafrost peatlands based on Fourier transform infrared (FTIR) spectroscopy. As a result, CUE was likely controlled by a low availability of easily degradable substrates at both geomorphic features, proposing that the cause for the difference in CUE lays elsewhere.

It could be proposed that the efficiency of the microbial community in the thermokarst was most strongly impacted by the change from in-situ anoxic to ex-situ oxic conditions. The microbial community in the thermokarst is adapted to the anoxic conditions found at-site. As the microbial community needs time to adapt to the sudden higher O_2 availability, it is possible that CUE was restricted in the thermokarst when conditions turned from anoxic to oxic (see chapter 4.4). This might have caused the lower CUE observed in the thermokarst compared to the palsa under oxic conditions.

Microbial growth was significantly lower in the thermokarst than in the palsa in depth section 5. MBC and microbial respiration did not show significant differences between the two

geomorphic features but tended to be lower in the thermokarst. Hence, the second part of hypothesis (1b) has to be rejected. MBC, microbial growth and respiration were likely hindered to a similar degree by an overall low availability of decomposable substrates (D'Andrilli et al., 2015). Similarly, Harris et al. (2023) did not find significant differences in respiration rates under oxic conditions between deep palsa and deep thermokarst peat as peat quality and degradability barely varied. The observed lower microbial growth in depth section 5 and the non-significantly lower values for biomass and respiration in the thermokarst were possibly caused by the change from anoxic to oxic conditions, as already proposed for CUE.

Anoxic

As discussed, peat quality and degradability did not differ between palsa and thermokarst, contrasting the expectation. Thus, the explanation previously made for the oxic conditions would suggest a lower microbial CUE in the palsa compared to the thermokarst when O_2 status changed from oxic in-situ conditions to the anoxic conditions created in the laboratory, but this was not the case here. Comparable to the oxic conditions, CUE under anoxic conditions tended to be higher in the permafrost palsa peat compared to equivalent thermokarst peat, and this was significant in depth section 4. This suggests that CUE can be higher in palsas compared to thermokarsts both under oxic as well as under anoxic conditions. Different factors, such as microbial mortality, could have caused a (short-term) enhancement of energy supply in the palsa when conditions turned from oxic to anoxic (see chapter 4.4), and this might have finally resulted in a higher CUE in the palsa compared to the thermokarst.

Respiration under anoxic conditions was higher in the thermokarst while MBC and microbial growth did not significantly differ between palsa and thermokarst but displayed overall higher values in the palsa. Estop-Aragonés et al. (2022) did not find differences in respiration under anoxic conditions between deep palsa and deep thermokarst peat, but Kirkwood et al. (2021) determined a higher respiration in a Canadian thermokarst compared to a palsa under anoxic conditions. The results derived here show that under anoxic conditions, microorganisms in the thermokarst react with a significantly higher respiration and more similar growth rates compared to the palsa. Under oxic conditions, respiration did not significantly differ but microbial growth was restricted as compared to the palsa. This again points to a better adaptation of microorganisms to anoxic conditions in the thermokarst.

4.3 Depth gradients of microbial parameters

4.3.1 Palsa

Hypothesis (2a) tested microbial parameters for differences between depth sections of the oxically incubated palsa peat. Here, the focus laid on differences between active layer and permafrost layer to unravel how the annual freezing and thawing of the active layer peat affects microbes in comparison to the freshly thawed peat from the deep permafrost layer. Thawing of the peat might release nutrients (Reyes & Lougheed, 2015; Finger et al., 2016) and available C (Nielsen et al., 2001; Dörsch et al., 2004; Osterkamp, 2007) for microbial use, and this could potentially influence microbial parameters (Matzner & Borken, 2008).

CUE did not significantly vary with depth in the palsa. A trend was observable with higher CUE values in depth sections 1 and 2 compared to depth sections 3 to 5. This observation disagrees with hypothesis (2a) which proposed a lower CUE in the active layer compared to the permafrost layer because of an anticipated higher N limitation in the active layer (Song et al., 2014; Reyes & Lougheed, 2015; Kjær et al., 2024). The lack of correlation of C/N ratio and N concentration with CUE might explain why CUE was not lower in the active layer compared to the permafrost layer. Overall, elemental concentrations and ratios varied little between active layer and permafrost layer. This could be a result of the epigenetic character of the studied palsa (see chapter 4.2.2). It could also be because the current permafrost layer might have gone through similar freeze-thaw cycles millions of years ago as the active layer today, resulting in similar peat characteristics in active layer and permafrost layer. This could explain why CUE did not significantly change along the depth gradient in the palsa.

The only significant difference along the palsa profile was a higher degradability of OM in the top depth sections, indicated by H/C ratio (D'Andrilli et al., 2015). Therefore, the correlation between H/C ratio and H concentration with CUE might explain why CUE displayed a decreasing trend with depth, although this trend was not significant (Blagodatskaya & Kuzyakov, 2008; Manzoni et al., 2012; Öquist et al., 2017). Based on the threshold proposed by D'Andrilli et al. (2015), OM in the studied palsa can be characterized as more easily degradable in the top two depth sections and as recalcitrant in the deeper depth sections. This result is similar to observations from Moore et al. (2018) who found non-significantly higher H/C ratios at top depths of Canadian peatlands. This could indicate that, although the vegetation found on palsas is rather scarce (Martin et al., 2019), the top depth sections were probably introduced to easily degradable C through fresh plant litter (Fuchsman, 1980; Rydin & Jeglum, 2013) and root exudates (Gunina & Kuzyakov, 2015). It further shows that the permafrost layer of the studied palsa does not host a large amount of degradable substrates for microbial use as decomposition is already advanced (see chapter 4.2.2). Hence, the tendency of a decreasing

CUE with depth can be attributed to a depletion of easily degradable substrates with peat age as indicated by H/C ratio and H concentration.

Similarly, MBC and microbial growth were highest in the top depth section. This result is to the most part in line with hypothesis (2a), but differences were not significant. Other studies reported similar patterns of MBC in permafrost peat (Song et al., 2014; Voigt et al., 2019). Lower temperatures in the deeper depth sections might have inhibited the activity of microorganisms as compared to the top (Jansson & Taş 2014; Altshuler et al., 2017). Moreover, the decreasing degradability of OM with depth might explain the observed pattern for MBC and growth. In the present study, MBC was positively correlated with O/C ratio (strong but not significant) and O content (significant). Comparable to the H/C ratio, the O/C ratio was found to act as a proxy for the degradability of OM in peat and soil (Zaccone et al., 2007; Zaccone et al., 2008; Moore et al., 2018; Zaccone et al., 2018). Easily degradable substrates, such as cellulose, display a higher O/C ratio than rather recalcitrant substrates, such as lignin, that are decomposing slower and remain in the peat for longer time (Moore et al., 2018). In the present study, the O/C ratio and O content showed relatively similar patterns as H/C ratio and H content with a decreasing tendency with depth. Similarly, other studies reported an increasing MBC with increasing degradability of OM (Fisk et al., 2003; Spohn et al., 2016a; Sihi et al., 2017; Jiang et al., 2022). Growth comparably increased with increasing degradability of OM, as can be seen from the correlation with H/C ratio and H concentration. This was already observed for mineral pasture and forest soils (Spohn et al., 2016a). Hence, a lower degradability of OM as well as lower temperatures in the permafrost might have caused the observed patterns of MBC and growth.

Contrasting hypothesis (2a), respiration was similar in the top depth section and the permafrost layer but lower in the deeper active layer. This is surprising because respiration is known to be affected by the availability of easily degradable C (Walz et al., 2017; Wang & Roulet, 2017; Harris et al., 2023; Kjær et al., 2024), and the indicators for degradability of C used in this study propose that substrate degradability is high throughout the active layer but low in the permafrost layer. On the other hand, respiration in the palsa was not found to be affected by any of the assessed elemental concentrations or ratios, including the used indicators for degradability of peat. It is possible that thawing of the permafrost layer released C that was not well reflected in H/C ratio or H concentration. For instance, microbial necromass was possibly released and decomposed by microorganisms after thaw (Herrmann & Witter, 2002; Dörsch et al., 2004). As microbial biomass only makes up a small portion of the peat (see chapter 4.1), it probably did not visibly affect H/C ratio and H concentration in the permafrost layer, but the release might have been sufficient for a short-term enhancement of respiration (Herrmann & Witter, 2002; Dörsch et al., 2004). It appears like under oxic conditions, this newly available C was primarily

invested into respiration. This is probably because of the comparably recalcitrant character of microbial necromass (Gunina & Kuzyakov, 2015) which increases the need for extracellular enzyme production and decreases the amount of energy available for growth (Manzoni et al., 2012). It could also show that in this system, respiration might react faster to changing conditions than MBC and growth. This could explain why respiration was significantly higher in depth section 5 than in depth section 2 while MBC and growth did not display a significant increase in the deeper depth sections.

The observations made for microbial respiration are in line with other studies. Kjær et al. (2024) concluded that after 96 days of oxic incubation, thawed permafrost peat from Áidejávri can display comparable respiration rates as peat from the top. Waldrop et al. (2010) observed no difference in respiration throughout their permafrost soil profiles in Alaska after 40 days of incubation. In the permafrost, more C was respired per unit MBC (qCO_2) than in the top depth section (Fig. A.3). As a result, respiration was similar in top depth section and permafrost layer, although MBC was lowest in the permafrost layer. As suggested by Waldrop et al. (2010), this shows that microbial abundance may be low, but there is still a relatively high potential for respiration in freshly thawed permafrost layers, resulting in similar or even higher total values compared to the active layer. This effect is probably only temporary. As C that was made available with thaw gets depleted, respiration in the permafrost layer can be expected to decrease to values more similar to those of the deeper active layer that is not regularly provided with degradable C (Semenchuck et al., 2016; Monteux et al., 2018).

4.3.2 Thermokarst

Hypothesis (2b) tested microbial parameters for differences along the depth gradient of the anoxically incubated thermokarst samples. The focus laid on differences between the newly formed peat in the top depth section and the peat of the deeper depth sections (former palsa peat) since here the most profound differences in peat composition, quality and overall conditions were expected.

CUE was lower in the newly formed peat than in the top of the former palsa peat but higher than in the bottom of the former palsa peat which reflects the separation of depth sections also shown in the PCA, but the differences were not significant. This is not fully in line with what was expected in hypothesis (2b). CUE in the thermokarst was proposed to be lower in the new peat than in the former palsa peat due to a larger N limitation (Oke & Hager, 2020; Harris et al., 2023). As discussed above, C/N ratio and N concentration were not correlated with CUE in the thermokarst, possibly because C/N ratio and N concentration did not differ strong enough with depth, and overall N availability was low (see chapter 4.2.2). This could explain why CUE was not significantly lower in the top depth section.

The correlation between H concentration and CUE indicates that CUE in the thermokarst increased with increasing peat degradability. Comparable to the palsa, OM in the top depth sections of the thermokarst can be described as more easily degradable while the lowest two depth sections consist of rather recalcitrant OM (D'Andrilli et al., 2015). Similarly, Harris et al. (2023) suggested a higher portion of more easily degradable C in newly formed thermokarst peat. This results from large input of plant debris due to an increased productivity caused by the comparably warm conditions in thermokarsts (Wilson et al., 2017; Taylor et al., 2019). Thus, although microbial activity in the thermokarst was restricted by the anoxic conditions (see chapter 4.2.1), a higher portion of degradable substrates in the top due to constant substrate input and younger age might have contributed to the overall but not significantly higher CUE values in the top depth sections compared to the lower depth sections (Blagodatskaya & Kuzyakov, 2008; Manzoni et al., 2012; Öquist et al., 2017).

The change in substrate degradability with depth, as indicated by H/C ratio and H concentration, could also explain the observed patterns for MBC and growth. Overall, MBC and microbial growth were low (see chapter 4.1), and methodology caused high variability (see chapter 4.5) which is why identifying differences with depth was difficult. Nonetheless, MBC and growth tended to be higher in the top than in the deeper depth sections. This is in line with hypothesis (2b). A higher MBC in the top depth sections was already observed in other peatland studies and can be ascribed to a decreasing degradability of substrates with depth (Parvin et al., 2018; Luo et al., 2022).

Respiration remained similar throughout the thermokarst profile. This is in contrast to what was anticipated in hypothesis (2b) and also contradicts the assumption of a lower degradability of peat in the deeper depth sections. Respiration was expected to be higher in the newly formed peat since there was more C readily available for microbial use and MBC was higher compared to the deeper depth sections (Wang et al., 2003; Harris et al., 2023). The results could show that respiration throughout the whole profile was primarily inhibited by the anoxic conditions (Turetsky, 2004; Song et al., 2014; Treat et al., 2014; Schädel et al., 2016; Estop-Aragonés et al., 2018a) and, therefore, differences in respiration along the depth gradient were not detectable, despite a higher proportion of available substrates at the top. This could further indicate a higher sensitivity of respiration than of MBC and microbial growth to anoxic conditions which could explain the different pattern observed for respiration compared to MBC and growth. This higher sensitivity to anoxic conditions can also be seen when comparing microbial parameters between palsa and thermokarst under different O_2 conditions (see chapter 4.2.1) and when comparing microbial parameters under oxic and anoxic conditions within the thermokarst (see chapter 4.4). The results derived here are in contrast to what other studies reported for thermokarsts as they found higher respiration rates in newly formed thermokarst

peat compared to peat from deeper depth sections (Harris et al., 2023; Kjær et al., 2024). Kjær et al. (2024) even performed their study in Áidejávri. However, the mentioned studies incubated their samples under oxic and not under anoxic conditions. This supports the assumption that respiration decreases with depth in thermokarsts due to decreasing peat degradability, but when conditions are anoxic, respiration might be strongly restricted throughout all depth sections, and differences along the depth gradient caused by variations in peat degradability might fade.

4.4 Microbial parameters under oxic vs. anoxic conditions

The third hypothesis tested differences in microbial parameters between oxic and anoxic conditions within the same geomorphic feature and the same depth section of the permafrost and former permafrost layer. Because of water table fluctuations, especially under extreme dry and wet weather conditions, the availability of O_2 can change within relatively short time scales. Climate change will support an increase of such extreme weather conditions (IPCC, 2023). The response to oxic and anoxic conditions might affect microorganisms in palsa and thermokarst. Moreover, when palsas collapse due to thermokarst, the previous oxic conditions change to anoxic (Zandt et al., 2020), and this might influence microbial parameters. Differences were only tested for the two deepest depth sections which likely host microbial communities that are better adapted to anoxic conditions (Wiersum, 1960; Kjær et al., 2024).

Palsa

Microbial CUE in the palsa was higher under anoxic than under oxic conditions, but this difference was only significant for depth section 4. This shows the tendency that under anoxic conditions less C has to be respired to achieve a similar growth rate as under oxic conditions. This is surprising since a lower CUE was anticipated for the anoxic conditions. Under anoxic conditions, substantial production of fermentation products and CH_4 with overall less energy gain for the microorganisms was expected as compared to oxic conditions (Conrad, 1999; Burgin et al., 2011). This was presumed to limit microbial CUE under anoxic conditions (Šantrůčková et al., 2004; Manzoni et al., 2012; Zheng et al., 2019).

Contrasting to the expectation, there was no or only very little net CH_4 production detected under anoxic conditions in the palsa. Palsas do not necessarily support high CH_4 emissions (Liebner et al., 2015). In the deep frozen layers of the palsa studied here, the low temperatures in the permafrost possibly reduced the abundance of methanogenic microorganisms (Yavitt et al., 2006; Helbig et al., 2016). Moreover, the low substrate quality possibly resulted in slow growth and little catabolism (see chapter 4.3.1). Therefore, CH_4 production was not elevated

after the relatively short anoxic incubation (Knoblauch et al., 2018), and there were no differences in respiration pathways between oxic and anoxic conditions.

The observation of a higher CUE under anoxic conditions could be explained in four ways. First, the change from in-situ oxic to ex-situ anoxic conditions in the palsa peat might have caused the death of aerobic bacteria and enhanced the amount of microbial necromass, increasing the energy supply for living microorganisms for a short time (Kästner et al., 2021; Shoemaker et al., 2021; Gunina & Kuzyakov, 2022). This could point to a short-term effect on microbes and microbial CUE that might occur when palsas collapse and change their O_2 conditions from oxic to anoxic. Yang et al. (2025) did not find a correlation between CUE and microbial necromass at the Loess Plateau in China, but in their study microbial communities were probably not as dependent on the supply of nutrients and C from microbial necromass as other sources were possibly already sufficient. Second, Adingo et al. (2021) state that a low O_2 availability does not necessarily result in a lowered CUE. Under oxic conditions, more of the gained energy has to be invested into processes other than growth. For instance, enhanced cell maintenance (Hoehler & Jørgensen, 2013) might be necessary because of e.g. oxidative cell damage. As a result, CUE could be similar or even lower under oxic conditions (Adingo et al., 2021). Third, the microbial community in the deep permafrost palsa sections could already be adapted to a lower O_2 availability to a certain degree (Wiersum, 1960; Kjær et al., 2024) and, therefore, might not largely react to varying O_2 conditions or might even need time to adjust to conditions with higher O_2 availability. Fourth, different incubation times of the oxic and anoxic samples might have caused the observed pattern of CUE (see chapter 4.5.2). In consequence, CUE under anoxic and oxic conditions in the palsa was possibly affected by one of these processes or by a combination of them.

MBC and microbial growth did not significantly differ between oxic and anoxic samples which is in contrast to what was anticipated in hypothesis (3). The results can possibly also be attributed to one of the processes mentioned above or to a combination of them. This not completely clear effect of O_2 condition on MBC was already observed in other peatland studies where samples were incubated for a maximum of 48 days (Song et al., 2014; Chari et al., 2021). It has to be kept in mind that methodology caused high variability which made it difficult to determine significant differences (see chapter 4.5), but overall, the results indicate that the change from oxic to anoxic conditions does not significantly restrict MBC and growth in palsas within ca. one month of pre-incubation.

Microbial respiration, on the other hand, was significantly lower when conditions were anoxic, supporting hypothesis (3). A reduced microbial respiration under anoxic conditions was observed in other permafrost soils as well (Turetsky, 2004; Song et al., 2014; Treat et al., 2014; Schädel et al., 2016; Estop-Aragonés et al., 2018a), and the effect is especially visible in

organic soils (Schuur et al., 2015). This shows that oxidative status has a significant effect on C loss in form of respiration, and this can be observed already after a relatively short incubation time. This could indicate that microbial respiration is inhibited by anoxic conditions stronger and faster than MBC and growth. It could further indicate that even if anoxic conditions caused the formation of microbial necromass as proposed above, respiration is restricted stronger by a low O_2 availability than it is enhanced by a short-term increase of available C. This is similar to what was proposed for the depth gradient in the thermokarst (see chapter 4.3.2). Hence, the change from oxic to anoxic conditions that happens when palsas collapse and form anoxic thermokarsts could support a rapid decrease in microbial respiration. This observation supports the conclusion made in other studies that the rate of respiration after permafrost thaw is highly dependent on the availability of O_2 (Schädel et al., 2016; Estop-Aragonés et al., 2018a; Estop-Aragonés et al., 2018b).

Thermokarst

Similar to the palsa, CUE in the thermokarst tended to be higher under anoxic conditions, but differences were not significant for both depth sections. In the thermokarst as well, there was no considerable CH_4 production observed under anoxic conditions. This is surprising since for thermokarsts, high CH_4 fluxes were measured both in-situ (Euskirchen et al., 2014; Liebner et al., 2015) as well as in anoxic incubation experiments (Hodgkins et al., 2014; Kirkwood et al., 2021). This is because in wet, anoxic thermokarsts, electron acceptors are depleted fast which supports methanogenesis (Wilson et al., 2019). The missing CH_4 production in the thermokarst studied here is possibly a cause of thermokarst age or inhibition of methanogenesis. CH_4 production typically is highest directly after permafrost collapse and decreases after the thermokarst has stabilized (Hugelius et al., 2020; Turetsky et al., 2020). This dynamic is possibly observed because right after permafrost collapse, trapped C suddenly becomes available, enhancing respiration processes, but then is depleted as decomposition advances (Semenchuck et al., 2016). Here, a mature thermokarst that most likely developed within the last century was studied which appears to not be prone to such an enhanced CH_4 production anymore. Moreover, a dominance of sedges rather than Sphagnum mosses was found to increase CH_4 production (Rooney-Varga et al., 2007; Olefeldt et al., 2012). This could play a role in the Sphagnum dominated thermokarst studied here. Hence, the overall conditions in the studied thermokarst do not seem to favor a high CH_4 production, and respiration pathways were the same under oxic and anoxic conditions.

Changes in O_2 condition hardly influenced CUE, MBC and microbial growth in deep sections of the thermokarst, suggesting a better adaptation of the microbial community to anoxic conditions and a slow adjustment to changing conditions. The high turnover times as well as the missing enhancement in CH_4 production even after 91 days of anoxic incubation additionally

support the assumption of a slow responsiveness of the microbial community in the thermokarst. Higher CO_2 production rates under oxic conditions, however, indicate an increasing catabolic metabolism associated with increasing mineralization of the peat (Kjær et al., 2024) already within a relatively short time frame.

4.5 Methodological limitations

4.5.1 Chloroform fumigation-extraction

A large difficulty in this study were the low or partially negative values determined for the MBC. Other studies already documented problems when determining MBC of peat with the CFE method. For example, Kaiser et al. (1992) concluded that OM has an effect on the determination of MBC as they observed a closer ratio between extractable C of fumigated and unfumigated peat samples compared to soils with less OM. This is similar to observations made by Sparling and West (1988). Hence, in the present study the negative values might be a result of the high OM content within the peat which makes it more difficult to determine differences between extracted C from fumigated and unfumigated samples and makes the process more sensitive to measuring errors. The high variability within the samples certainly also plays a role and might have led to the reported negative values. Therefore, it was necessary to find a solution on how to deal with the negative MBC values.

Since it can be assumed that in every soil there is at least a minimum amount of microbial biomass, all negative values were set to a minimum value. Studies performed in the active layers of permafrost soils report a lowest MBC of around 100 µg C g $soil^{-1}$ (Shang et al., 2016; Dong et al., 2023). Low MBC values of non-permafrost peat soils were reported by Luo et al. (2022) who measured an MBC of around 12.0 µg C g $peat^{-1}$ in 80-100 cm depth of a heavily grazed Chinese peatland during the growing season. Fisk (2003) and Parvin et al. (2018) measured lowest MBC of around 60.0 µg C g $peat^{-1}$ in peat soils. Based on these considerations, a minimum value for MBC of 50.0 µg C g $peat^{-1}$ was set.

4.5.2 DNA extraction and carbon use efficiency

Difficulties were also encountered during DNA extractions, pointing to limitations of the method when working with permafrost peat. First of all, DNA extractions were difficult because humic acids were extracted together with the anticipated DNA (Tsai & Olson, 1991; Tsai & Olson, 1992; Tebbe & Vahjen, 1993). Peat with its high content of OM (IUSS Working Group, 2022) might be especially susceptible to this kind of contamination of extracted DNA. This is why the extracted DNA had to be purified in a follow-up step (Tsai & Olson, 1992; Tebbe & Vahjen, 1993) before performing the ^{18}O measurements. However, in the twelve samples for which

DNA concentrations were determined again after purification (see chapter 2.7), purification led to a DNA loss between 12.0% and 84.3% with an average of 61.3% (Tab. A.2). Additionally, it did not completely remove the humic acids as some of the purified samples were still not completely transparent, pointing to a rest-contamination. This could have negatively affected the ^{18}O measurements.

Moreover, DNA concentrations determined in this study were partially extraordinary low (Fig. A.5). Low DNA quantities were also reported in another study working with permafrost soils, which complicated the following analysis (Graham et al., 2012). In the case of the present study, the low DNA concentrations might have complicated the ^{18}O measurements, especially since they decreased even more after the purification as discussed above. This is possibly reflected by two samples with a negative ^{18}O enrichment value.

Finally, the mentioned difficulties also affected the results for CUE as both MBC and DNA concentration are substantial parts of the CUE calculation (see eq. 5-7), and they may have impacted the results. For instance, some of the CUEs determined in the present study for single replicates reached values higher than 0.90. These very high values collide with literature as thermodynamic constraints (Roels, 1980) were documented to hinder microbes from reaching CUEs higher than 0.85 (Gommers et al., 1988; Geyer et al., 2016).

Furthermore, determination of CUE under oxic and anoxic conditions was possibly affected by the methodological design. Niedźwiecka et al. (2025) determined the CUE in spruce forest soils using ^{13}C and observed a higher CUE under oxic conditions in the beginning of the incubation period. However, after 50 hours of incubation, the CUE under anoxic conditions exceeded the one under oxic conditions. In the present study, incubation times for both microbial respiration and incorporation of $^{18}H_2O$ into the microbial biomass (microbial growth) were close to the 50 hours that marked the tipping point in the study of Niedźwiecka et al. (2025) and generally higher for the anoxic than the oxic samples (see chapter 2.5 and 2.7). Consequently, the higher CUE in the anoxic samples might have been a result of methodology.

5 Conclusion and outlook

Throughout all studied peat samples, microbial biomass and activity where highest in the top 20 cm of the palsa under oxic conditions which could suggest that active palsa layers release a slightly enhanced amount of C into the atmosphere during the growing season. The strong degradation of the permafrost palsa peat as indicated by H/C ratio and H concentration, however, suggests a low potential for C release with thawing of deeper palsa sections (i.e. active layer deepening or thermokarst). Especially if palsas collapse to thermokarsts and conditions change from oxic to anoxic, no enhanced C release should be expected. The strong degradation of peat as well as the prevailing anoxic conditions that restrict microbial activity propose that thermokarsts do not significantly contribute to climate warming. Low microbial biomass and activity under anoxic and oxic conditions that was detected in the thermokarst below 120 cm depth as well as a slow responsiveness of the microbial community further suggest little potential for C losses by peat decomposition at fluctuating water table. Thus, the thawing of epigenetic palsas as a result of climate warming might have only little potential for positive climate feedback.

Looking forward, further studies could be performed in different thermokarst systems with distinct vegetation cover. It could be investigated how dry spells with low water levels in thermokarsts affect microbial parameters and how the microbial parameters relate to the C cycle. This could allow statements about whether these systems gain or lose C, especially under changing climatic conditions.

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Appendix

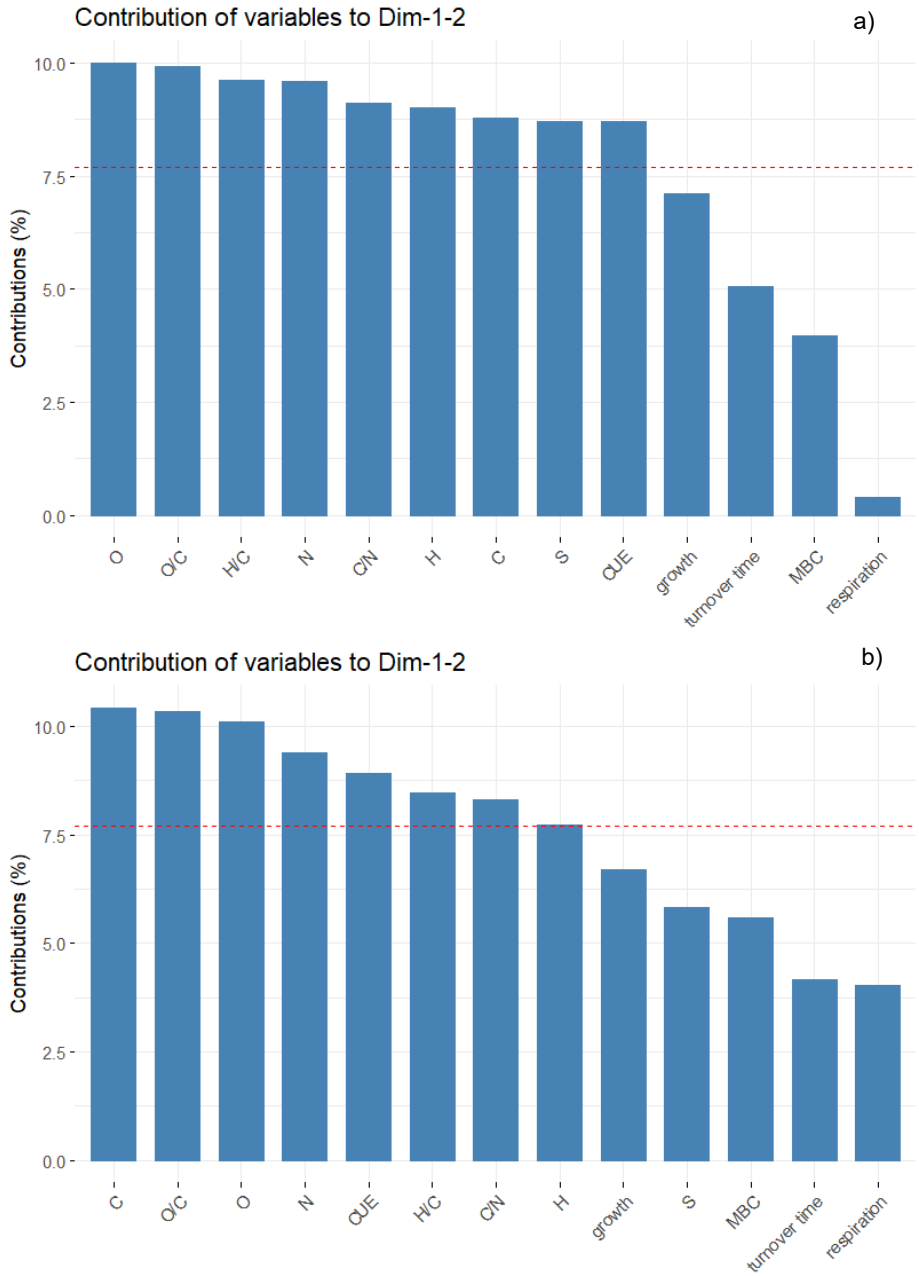


Figure A. 1: *Contribution of variables to the first two dimensions of the principal component analysis (PCA) a) for a Norwegian palsa and b) for a Norwegian thermokarst.*

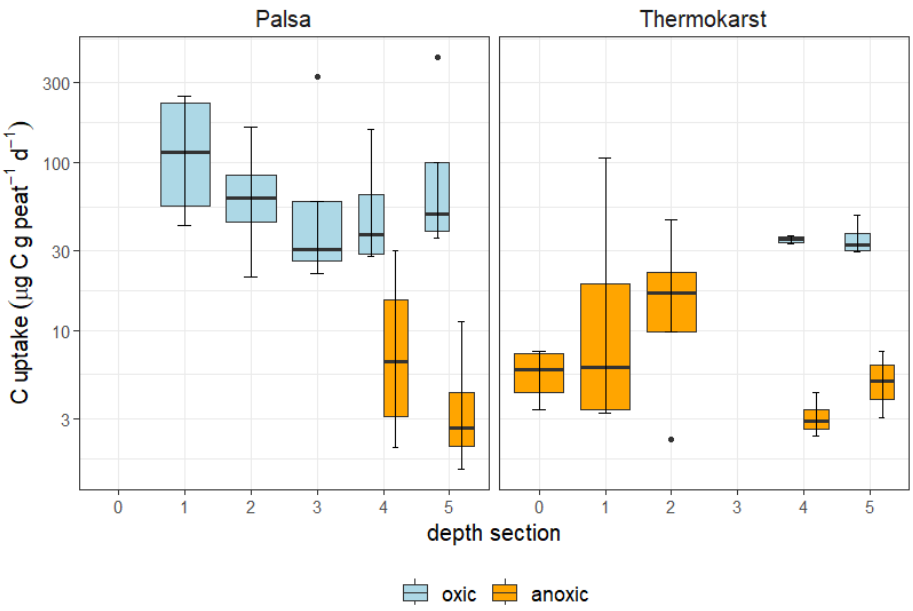


Figure A. 2: *Boxplot showing the microbial carbon uptake (C uptake), calculated from the sum of microbial growth and respiration, in a Norwegian palsa and thermokarst in different depth sections (0-5) and at different oxygen conditions (oxic, anoxic).*

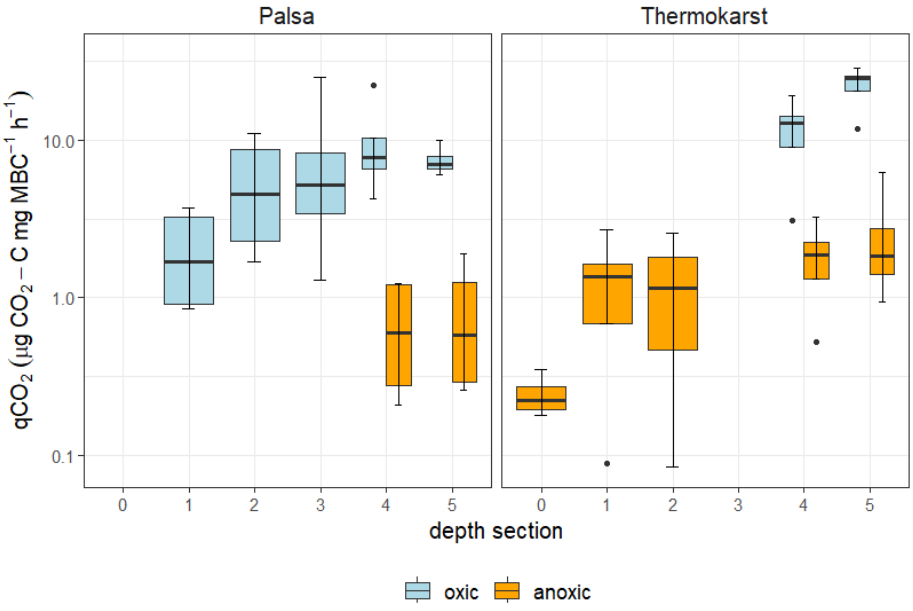


Figure A. 3: *Boxplot showing microbial respiration per unit microbial biomass carbon (qCO_2) in a Norwegian palsa and thermokarst in different depth sections (0-5) and at different oxygen conditions (oxic, anoxic).*

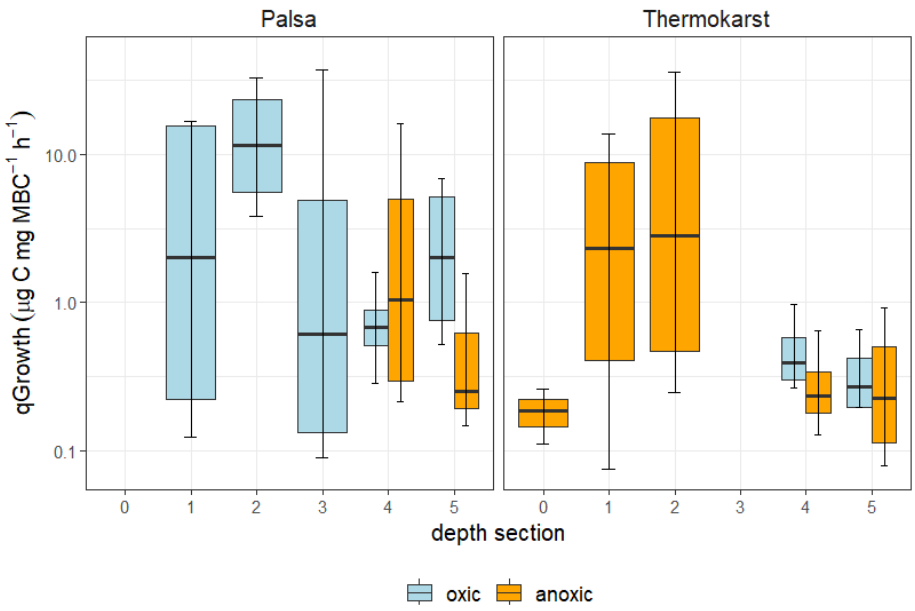


Figure A. 4: *Boxplot showing microbial growth per unit microbial biomass carbon (qGrowth) in a Norwegian palsa and thermokarst in different depth sections (0-5) and at different oxygen conditions (oxic, anoxic).*

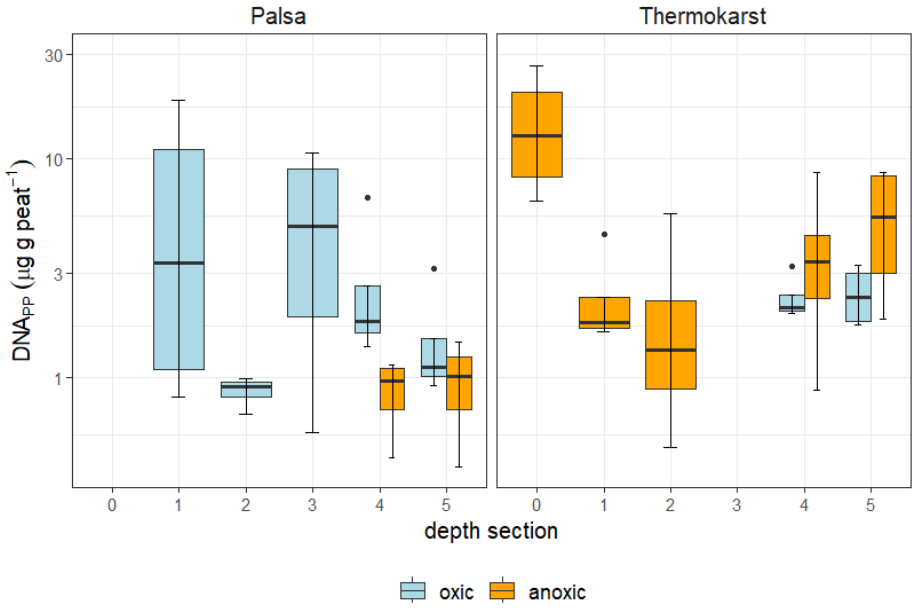


Figure A. 5: *Boxplot showing DNA concentrations determined with a microplate reader using PicoGreen after extraction with the SPINeasy™ DNA Plant Kit (MP Biomedicals; DNA$_{PP}$) in a Norwegian palsa and thermokarst in different depth sections (0-5) and at different oxygen conditions (oxic, anoxic).*

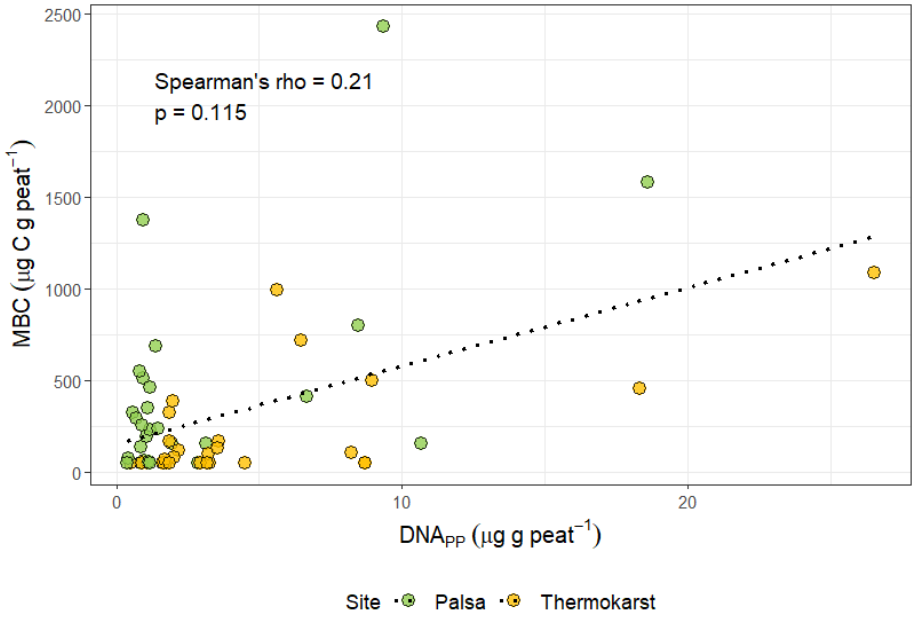


Figure A. 6: *Linear DNA concentration determined with a microplate reader using PicoGreen after extraction with the SPINeasy™ DNA Plant Kit (MP Biomedicals; DNA_{PP}) and microbial biomass carbon (MBC) in a Norwegian palsa and thermokarst.*

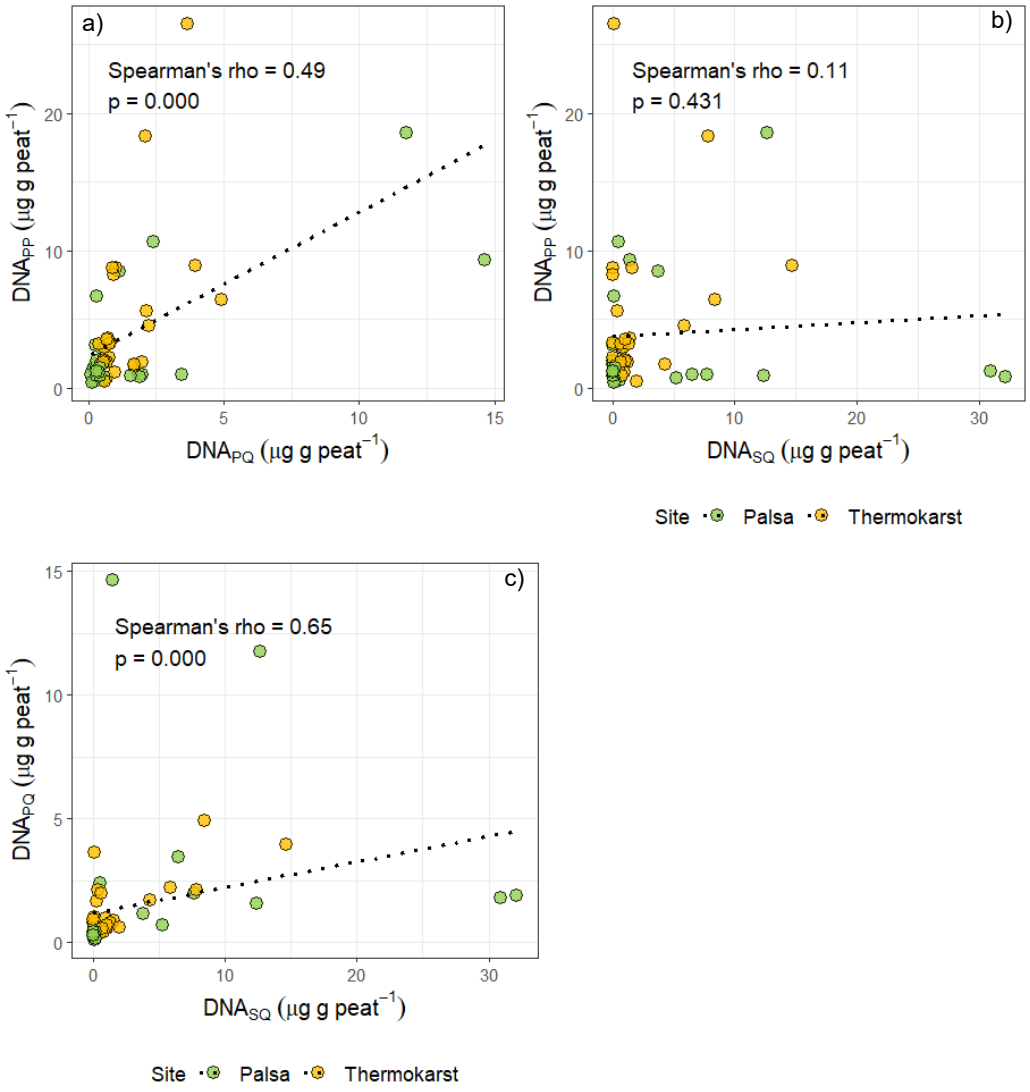


Figure A. 7: *Linear correlation of a) DNA concentration determined with a Qubit after extraction with the SPINeasy™ DNA Plant Kit (MP Biomedicals; DNA_{PQ}) and DNA concentration determined with a microplate reader using PicoGreen after extraction with the SPINeasy™ DNA Plant Kit (MP Biomedicals; DNA_{PP}), b) DNA concentration determined with a Qubit after extraction with the SPINeasy™ DNA Soil Kit (MP Biomedicals; DNA_{SQ}) and DNA concentration determined with a microplate reader using PicoGreen after extraction with the SPINeasy™ DNA Plant Kit (MP Biomedicals; DNA_{PP}) and c) DNA concentration determined with a Qubit after extraction with the SPINeasy™ DNA Soil Kit (MP Biomedicals; DNA_{SQ}) and DNA concentration determined with a Qubit after extraction with the SPINeasy™ DNA Plant Kit (MP Biomedicals; DNA_{PQ}) in a Norwegian palsa and thermokarst.*

Table A. 1: *Microbial biomass carbon (MBC) at four plots of a Norwegian palsa (AP1-4) and four plots of a Norwegian thermokarst (AT1-3, AT5) at different oxygen conditions (oxic, anoxic) and in different depth sections (0-5). Values that were corrected to the minimum value of 50.0 µg C g peat^{-1} are indicated in italics.*

Plot	**Oxygen condition**	**Depth section**					
		0 (µg C g peat^{-1})	**1** (µg C g peat^{-1})	**2** (µg C g peat^{-1})	**3** (µg C g peat^{-1})	**4** (µg C g peat^{-1})	**5** (µg C g peat^{-1})
AP1	oxic	-	1,580	509	326	684	1,371
AP2	oxic	-	458	58.8	798	411	190
AP3	oxic	-	2,430	289	*-71.9*	*-63.6*	228
AP4	oxic	-	547	*-44.9*	158	154	155
AT1	oxic	-	-	-	-	115	*17.2*
AT2	oxic	-	-	-	-	79.0	164
AT3	oxic	-	-	-	-	386	*27.7*
AT5	oxic	-	-	-	-	101	*28.1*
AP1	anoxic	-	-	-	-	71.9	*-255*
AP2	anoxic	-	-	-	-	348	233
AP3	anoxic	-	-	-	-	139	257
AP4	anoxic	-	-	-	-	53.2	*-137*
AT1	anoxic	1,088	*-14.8*	995	-	*-84.0*	103
AT2	anoxic	501	321	*-25.8*	-	170	*30.2*
AT3	anoxic	452	51.1	*-6.89*	-	*-18.5*	131
AT5	anoxic	716	65.8	*-113*	-	*-64.9*	*6.24*

Table A. 2: *DNA concentration determined with a microplate reader using PicoGreen before purification and after purification as well as the DNA lost in the purification step at two plots of a Norwegian palsa (AP1, AP2) and one plot of a Norwegian thermokarst (AT2) and in two depth sections (1, 2); [1] = sample labelled with ^{16}O, [2] = sample labelled with ^{18}O.*

Plot	**Depth section**	**DNA concentration before purification** (ng μl^{-1})	**DNA concentration after purification** (ng μl^{-1})	**DNA lost** (%)
AP2[1]	1	1.83	0.52	71.5
AP2[2]	1	1.91	0.66	65.4
AP2[1]	2	1.68	0.46	72.7
AP2[2]	2	1.36	0.62	54.3
AP4[1]	1	1.16	0.98	15.6
AP4[2]	1	1.30	1.14	12.0
AP4[1]	2	1.30	0.48	62.7
AP4[2]	2	1.33	0.51	61.4
AT2[1]	1	2.25	0.35	84.3
AT2[2]	1	2.13	0.34	83.9
AT2[1]	2	1.24	0.30	75.5
AT2[2]	2	1.20	0.28	76.7

Table A. 3: *Results from the statistical tests of elemental concentrations in different depth sections (0-5) of a Norwegian palsa and thermokarst.*

	C			N			O			H			S		
	df	F-value	p-value	df	F-value	p-value	df	F-value	p-value	df	F-value	p-value	df	F-value	p-value
Palsa	4, 15	0.119	0.974	4, 15	1.951	0.154	4, 15	0.897	0.490	4, 15	1.700	0.202	4, 15	3.696	**0.028**
1-2													15	-0.800	0.927
1-3													15	-2.507	0.141
1-4													15	-2.824	0.081
1-5													15	-3.070	0.051
2-3													15	-1.707	0.459
2-4													15	-2.024	0.302
2-5													15	-2.270	0.208
3-4													15	-0.317	0.998
3-5													15	-0.563	0.979
4-5													15	-0.246	0.999
Thermokarst	4, 15	13.06	**< 0.0001**	4, 15	16.18	**< 0.0001**	4, 15	14.50	**< 0.0001**	4, 15	14.75	**< 0.0001**	4, 15	3.260	**0.041**
0-1	15	-5.447	**< 0.001**	15	-6.052	**< 0.001**	15	5.921	**< 0.0001**	15	-3.045	0.054	15	-1.984	0.319
0-2	15	-6.72	**< 0.001**	15	-7.585	**< 0.0001**	15	6.968	**< 0.0001**	15	-2.362	0.180	15	-2.081	0.278
0-4	15	-4.93	**0.002**	15	-5.061	**0.001**	15	5.161	**< 0.001**	15	2.835	0.080	15	-3.468	**0.024**
0-5	15	-4.562	**0.003**	15	-4.598	**0.003**	15	5.022	**0.001**	15	2.572	0.126	15	-2.601	0.12
1-2	15	-1.273	0.710	15	-1.533	0.559	15	1.047	0.830	15	0.682	0.957	15	-0.096	1.000
1-4	15	0.517	0.984	15	0.991	0.855	15	-0.761	0.938	15	5.879	**< 0.001**	15	-1.484	0.587
1-5	15	0.885	0.898	15	1.454	0.605	15	-0.899	0.893	15	5.617	**< 0.001**	15	-0.617	0.970
2-4	15	1.790	0.414	15	2.524	0.137	15	-1.808	0.405	15	5.197	**< 0.001**	15	-1.387	0.644
2-5	15	2.158	0.247	15	2.986	0.060	15	-1.946	0.337	15	4.934	**0.001**	15	-0.520	0.984
4-5	15	0.368	0.996	15	0.462	0.990	15	-0.139	1.000	15	-0.262	0.999	15	0.867	0.905
Palsa vs. Thermokarst	df	t	p-value	df	t	p-value	df	t	p-value	df	t	p-value	df	t	p-value
1-1	6	-0.567	0.592	6	-1.975	0.096	6	1.447	0.198	6	-1.183	0.279	6	-3.451	**0.014**
2-2	3.481	-1.736	0.168	6	-1.781	0.125	3.331	1.795	0.161	6	-1.103	0.312	6	-1.626	0.155
4-4	3.340	0.111	0.918	3.323	0.907	0.426	3.52	0.392	0.717	6	1.505	0.183	6	-0.001	0.999
5-5	6	0.619	0.559	3.534	2.095	0.113	6	-0.058	0.956	6	0.654	0.538	6	0.693	0.514

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Table A. 4: *Results from the statistical tests of stoichiometric ratios in different depth sections (0-5) of a Norwegian palsa and thermokarst.*

	C/N			H/C			O/C		
	df	F-value	p-value	df	F-value	p-value	df	F-value	p-value
Palsa	4, 15	2.076	0.135	4, 15	7.158	**0.002**	4, 15	0.612	0.661
1-2				15	-0.129	1.000			
1-3				15	2.717	0.098			
1-4				15	3.558	**0.021**			
1-5				15	3.688	**0.016**			
2-3				15	2.846	0.078			
2-4				15	3.686	**0.016**			
2-5				15	3.817	**0.012**			
3-4				15	0.841	0.914			
3-5				15	0.971	0.864			
4-5				15	0.131	1.000			
Thermo-karst	4, 15	13.91	**< 0.001**	4, 15	66.26	**< 0.001**	4, 15	15.81	**< 0.001**
0-1	15	5.448	**< 0.001**	15	6.650	**< 0.001**	15	6.092	**< 0.001**
0-2	15	7.100	**< 0.001**	15	9.983	**< 0.001**	15	7.266	**< 0.001**
0-4	15	4.188	**0.006**	15	14.26	**< 0.001**	15	5.552	**< 0.001**
0-5	15	3.658	**0.017**	15	13.19	**< 0.001**	15	5.329	**< 0.001**
1-2	15	1.652	0.489	15	3.333	**0.032**	15	1.174	0.766
1-4	15	-1.259	0.718	15	7.611	**< 0.001**	15	-0.540	0.982
1-5	15	-1.790	0.414	15	6.536	**< 0.001**	15	-0.763	0.937
2-4	15	-2.911	0.069	15	4.278	**0.005**	15	-1.714	0.455
2-5	15	-3.442	**0.026**	15	3.203	**0.040**	15	-1.937	0.341
4-5	15	-0.530	0.983	15	-1.075	0.816	15	-0.223	0.999
Palsa vs. Thermo-karst	df	t	p-value	df	t	p-value	df	t	p-value
1-1	6	2.132	0.077	4.227	-1.439	0.220	6	1.109	0.310
2-2	3.046	1.877	0.156	6	1.362	0.222	3.306	1.762	0.168
4-4	3.291	-1.072	0.356	6	1.878	0.110	3.437	0.237	0.826
5-5	3.290	-2.149	0.113	6	0.331	0.752	6	-0.300	0.774

Table A. 5: *CH_4 production at four plots of a Norwegian palsa (AP1-4) and four plots of a Norwegian thermokarst (AT1-3, AT5) in different depth sections (0-5) after oxic incubation for 48 hours.*

Plot	**Depth section**					
	0 (ppm)	**1** (ppm)	**2** (ppm)	**3** (ppm)	**4** (ppm)	**5** (ppm)
AP1	-	4.14	1.22	1.87	1.87	1.77
AP2	-	0.00	1.71	1.74	1.84	1.82
AP3	-	0.26	1.38	1.75	1.73	1.75
AP4	-	0.79	1.47	1.00	1.75	1.76
AT1	-	-	-	-	1.77	1.74
AT2	-	-	-	-	1.88	1.85
AT3	-	-	-	-	1.90	1.79
AT5	-	-	-	-	1.78	1.76

Table A. 6: *CH_4 production at four plots of a Norwegian palsa (AP1-4) and four plots of a Norwegian thermokarst (AT1-3, AT5) in different depth sections (0-5) after anoxic incubation for 160 hours*

Plot	**Depth section**					
	0 (ppm)	**1** (ppm)	**2** (ppm)	**3** (ppm)	**4** (ppm)	**5** (ppm)
AP1	-	-	-	-	0.00	0.00
AP2	-	-	-	-	0.13	0.00
AP3	-	-	-	-	0.08	0.13
AP4	-	-	-	-	0.00	0.11
AT1	1.35	1.02	0.50	-	0.33	0.24
AT2	0.10	0.53	0.27	-	0.09	0.38
AT3	0.15	0.26	0.35	-	0.12	0.45
AT5	0.52	0.11	0.44	-	0.12	0.21

Table A. 7: *CH_4 production at four plots of a Norwegian thermokarst (AT1-3, AT5) in different depth sections (0, 1, 2, 4, 5) after long-term anoxic incubation for 46 (t1) and 91 (t2) days.*

Plot	**Depth section**	**t1** (ppm)	**t2** (ppm)
AT1	0	0.48	0.99
AT1	1	1.36	40.8
AT1	2	24.3	77.4
AT1	5	0.15	2.41
AT2	0	0.45	0.54
AT2	4	2.13	0.43
AT3	1	1.83	47.3
AT3	2	0.88	2.65
AT5	2	1.75	2.74
AT5	4	0.99	0.65

Table A. 8: *Spearman correlation coefficients of microbial parameters (microbial carbon use efficiency (CUE), microbial biomass carbon (MBC), microbial growth, microbial respiration and turnover time of microbial biomass) and elemental concentrations and ratios in a Norwegian palsa and thermokarst (Thermo). Significant correlations ($p < 0.05$) are indicated in bold.*

Element	**CUE**		**MBC**		**Growth**		**Respiration**		**Turnover time**	
	Palsa	Thermo	Palsa	Thermo	Palsa	Thermo	Palsa	Thermo	Palsa	Thermo
C	0.17	0.37	-0.44	-0.31	0.18	0.30	0.03	**-0.63**	**-0.54**	**-0.50**
N	0.12	0.16	-0.27	-0.44	0.19	0.07	0.15	-0.43	-0.43	-0.37
H	**0.65**	**0.46**	-0.03	0.16	**0.65**	0.45	-0.04	-0.36	**-0.73**	-0.26
O	-0.21	-0.27	**0.45**	0.36	-0.21	-0.19	0.02	**0.54**	**0.55**	0.38
S	**-0.59**	-0.22	**-0.49**	**-0.52**	**-0.63**	-0.36	-0.09	-0.13	0.33	-0.16
C/N	-0.08	-0.13	0.27	0.45	-0.14	-0.05	-0.15	0.41	0.38	0.35
H/C	**0.60**	0.28	0.22	**0.50**	**0.54**	0.32	-0.27	0.06	-0.38	0.10
O/C	-0.25	-0.29	0.43	0.31	-0.25	-0.21	-0.03	**0.58**	**0.59**	0.41

Table A. 9: *Spearman correlation coefficients of microbial parameters (microbial carbon use efficiency (CUE), microbial biomass carbon (MBC), microbial growth, microbial respiration and turnover time of microbial biomass) in a Norwegian palsa and thermokarst (Thermo). Significant correlations (p < 0.05) are indicated in bold.*

Parameter	**CUE**		**MBC**		**Growth**		**Respiration**		**Turnover time**	
	Palsa	Thermo	Palsa	Thermo	Palsa	Thermo	Palsa	Thermo	Palsa	Thermo
CUE										
MBC	0.18	0.32								
Growth	**0.75**	**0.80**	**0.46**	**0.49**						
Respiration	-0.24	**-0.76**	**0.56**	0.07	**0.40**	-0.32				
Turnover time	**-0.70**	**-0.54**	0.06	0.08	**-0.84**	**-0.73**	-0.17	0.24		

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Declaration of Authenticity

I, Mariele Castan, matriculation number 1920930, hereby confirm that the present thesis is solely my own work and that I have used no other than the stated sources and aids. All parts that are taken in word or sense from one of these sources are marked with a reference to the source. This thesis has not been used for the award of any other academic degree.

Bayreuth, 15.08.2025



Mariele Castan